

From experimental autonomy to scientific understanding in programmable chemical worlds

Jiangjie Qiu^{1,*}, Yijun Li^{1,*}, Yaotian Yang^{1,*}
Honghao Chen¹, Wentao Li¹, Xiaonan Wang^{1,†}

¹Beijing Key Laboratory of Artificial Intelligence for Advanced Chemical Engineering Materials, State Key Laboratory of Chemical Engineering and Low-Carbon Technology, Department of Chemical Engineering, Tsinghua University, Beijing 100084, China

*Jiangjie Qiu, Yijun Li and Yaotian Yang contributed equally.

†Correspondence: wangxiaonan@tsinghua.edu.cn

Abstract

Self-driving laboratories increasingly design procedures and pursue research paths. Equipment and workflow constraints, experimental costs, safety requirements and unknown mechanisms make their scientific learning difficult to study systematically. We introduce ChemWorld, programmable chemical worlds where agents compose experiments on evolving samples while evaluators independently control process laws and supplied information. Using one agent configuration across 240 campaigns in six system families, we connect autonomous research histories to operational outcomes and sealed predictions under new interventions. Agents adapt procedures and can benefit from resources and knowledge, but operational and predictive gains need not coincide. Correct partial prior information improves equilibrium predictions in one regime but increases error 8.7-fold under extreme dilution relative to withholding that information. Forecasts preserve a plateau where equilibrium responses change, yet depart from near-constant crystallization purity. These findings highlight judging the scope of experimentally acquired knowledge as a central challenge in translating experimental autonomy into scientific understanding.

Self-driving laboratories are extending the reach of experimental science through automated execution, continuous feedback and efficient searches for useful conditions. Language-model agents further connect scientific instructions, tools and laboratory operations, opening possibilities for more flexible research workflows¹⁻³. As systems gain the freedom to choose operation sequences, intermediate measurements and subsequent experiments, they increasingly shape their own research paths. The scientific opportunity extends from executing procedures and optimizing conditions to deciding which evidence to acquire and what that evidence implies.

Greater experimental freedom also makes scientific learning harder to evaluate. Different research paths can support the same operating success while exposing different aspects of a chemical system. A productive recipe may follow from a transferable explanation or from a locally effective empirical rule. Physical laboratories offer rich possibilities for discovery, while each implemented workflow has practical limits imposed by equipment, materials and procedures. Unknown mechanisms, evolving material states and the cost of repeated experiments complicate controlled comparisons of how autonomous researchers acquire and use evidence. Understanding their progress therefore requires connecting experimental achievements to the histories and judgments that produced them.

We examine an observable component of scientific understanding: how autonomous researchers use evidence they acquire to predict outcomes under new interventions. Research histories, public scientific accounts and sealed predictions connect experimental achievement to judgments about where relationships remain applicable. Prediction is one window onto this capacity; it does not by itself establish complete mechanistic understanding.

ChemWorld provides a computational setting for this investigation. Optimization benchmarks offer repeatable objectives^{4,5}, and interactive chemistry and discovery environments support multi-step experimentation and knowledge assessment⁶⁻⁸. ChemWorld combines composable chemical processes, persistent experimental state and flexible operations with evaluator control over world and information conditions. The

researcher can organize experiments within executable chemical domains; the evaluator can configure compatible worlds and compare research conditions while retaining their complete operational histories. This combination makes autonomous evidence acquisition and its subsequent use accessible to controlled study.

We examine 240 campaigns across six chemical system families, following the original researcher from experiment selection to operating recommendations, scientific accounts and sealed predictions. Agents develop adaptive procedures and benefit from additional resources and relevant knowledge, but operating and predictive improvements do not consistently coincide. More strikingly, correct prior information reverses its predictive benefit across equilibrium response regimes, whereas crystallization forecasts unnecessarily depart from stable purity despite predicting recovery effectively. These findings motivate applicability judgments as a critical dimension of scientific learning: determining when an observed relationship should persist, when it should change and which experiments could resolve the difference.

Results

Tracing autonomous experimentation in controlled chemical worlds

ChemWorld constructs executable worlds from compatible process and observation components. Operations change composition, phase allocation, thermal history and remaining material, whereas measurements expose only declared quantities. Operations act on persistent samples, and their history determines the state available for subsequent measurements and decisions. Researchers choose valid procedures, repetitions and intermediate measurements within chemical and resource constraints; evaluators independently configure compatible process laws, private parameters and supplied information. A world can therefore change behind a common public interface, or remain fixed while the researcher’s information changes. Because observations arise from the experiments actually chosen, supplied information can affect both evidence acquisition and interpretation (Fig. 1). State transitions, resource accounting and replay are qualified separately (Methods).

The study comprises 240 campaigns in five-world, three-arm blocks. Opaque provides usable public experimental contracts and anonymous identifiers, Aligned adds correct partial nominal information and MisIndexed supplies a fixed mismatch or reassignment. The same agent configuration conducts each campaign; budget comparisons use independent 12- and 24-batch sessions in the systems with both envelopes. Where required, a sealed recommendation receives an independent retest. The original session then provides a scientific account (K1), predictions for withheld conditions (Q) and retrospective evidence review (K2), retaining its research history. Neither Q nor K2 receives reference feedback, and K2 cannot revise Q. All campaigns remain included, including two with incomplete final-assay budgets.

A twelve-batch crystallization campaign shows this traceability in action (Fig. 2). The first four batches fail the particle-quality constraint with 99.5-100.0% fines. Batch 5 changes the process to controlled cooling before seeding and yields the first feasible result, with 30.3% fines and 40.0% recovery. Further adjustments raise feasible recovery to 45.5% in batch 8; its sealed recipe returns 43.9% recovery and 28.4% fines on independent retest. A later solvent reduction in batch 11 again violates the quality constraint, with 72.6% fines. The trajectory connects repeated failure, procedural adaptation, local improvement and renewed constraint violation to an externally evaluated outcome. Several conditions changed together, so the contribution of omitting quench is not identified. Questions shown in Fig. 2 are author reconstructions from recorded actions, and the nucleation-history explanation was written after the experiments rather than recorded as contemporaneous reasoning.

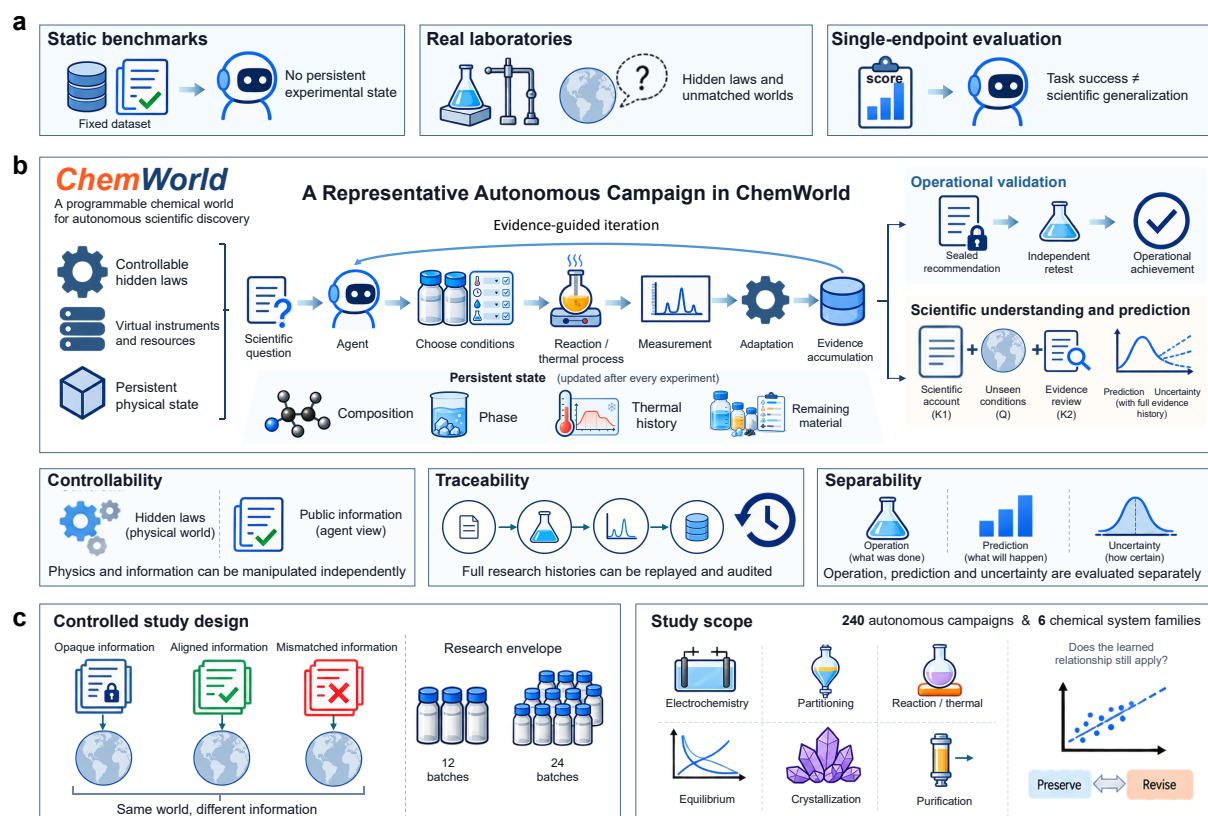


Figure 1: ChemWorld as a controlled instrument for autonomous scientific research. **a**, Evaluation challenges posed by fixed datasets, physical laboratories and single-endpoint task scores. **b**, An autonomous campaign connects experimental choice, process execution, measurement and evidence-guided adaptation in a persistent physical world. Hidden laws and public information can be varied independently, research histories can be replayed, and operation, prediction and uncertainty are assessed separately. Where required, a recommendation is sealed and independently retested. The original session then provides a scientific account (K1), sealed predictions (Q) and retrospective evidence review (K2), in that order with its research history available. Neither Q nor K2 receives target feedback. **c**, Matched-world information conditions, independent 12- and 24-batch research envelopes, and the completed study's 240 campaigns across six system families. "Mismatched information" denotes MisIndexed. Budget comparisons apply to the systems with both research envelopes.

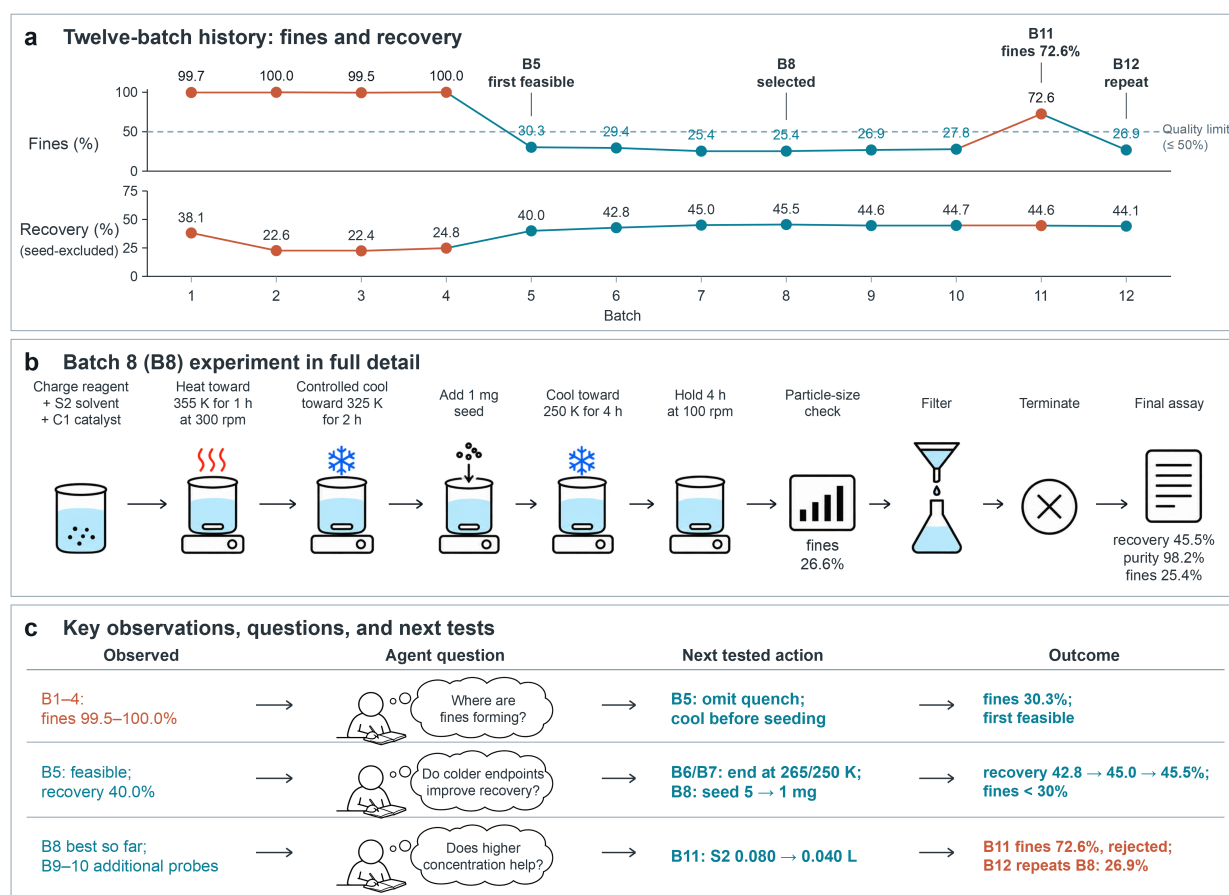


Figure 2: One complete autonomous crystallization research session. a, All twelve source batches in the retrospectively selected W04 MisIndexed session. Values are recorded final assays; teal marks quality-feasible batches and ochre marks failures. The dashed line is the 50% fines limit. Batch 5 is the first feasible result; batch 8 is the eventual sealed recommendation, batch 11 fails the fines rule and batch 12 repeats the selected recipe. b, All twelve operation calls of source batch 8, grouped into ten visible stages. The intermediate particle-size check reads 26.6% fines; the final assay reads 25.4%. Temperatures are requested targets, not measured trajectories. The independent retest uses the sealed batch-8 recipe with a different observation seed. c, Three sequences of observations, subsequent actions and outcomes. The questions are author reconstructions from recorded actions, not contemporaneous model quotations; K1 was written after the experiment series. Batch 5 changed several process conditions together, so the role of omitting quench is not identified. Recovery excludes seed mass. Feasibility requires purity at least 80%, fines at most 50% and particles present; fines are particles smaller than 20 micrometres. This selected case illustrates the research process, not a cohort frequency.

Operational and predictive gains need not coincide

Research commissions change priorities without prescribing experimental paths. Discovery prioritizes developing and testing an explanatory, predictive account, whereas optimization prioritizes a high-scoring procedure. We compare independent campaigns matched within world, information condition, budget and prior locus, evaluating recommendation retests and withheld predictions separately (Fig. 3). Optimization has the higher retest score in 26 of 30 electrochemical pairs; thirteen of these pairs also have lower score-prediction mean absolute error (MAE), whereas thirteen have higher MAE. In reaction processing, discovery performs better on both endpoints in seventeen of thirty pairs. Thus neither commission has a predetermined overall advantage, and operating and predictive outcomes require separate evaluation.

A selected electrochemical pair makes this separation concrete (Fig. 3c,d). Discovery compares materials and conditions before extending duration and increasing the current cap. Optimization refines a promising solvent-electrolyte combination, raising its observed score from 0.675 in batch 3 to 0.749 in batch 12. Recommendation retests are 0.718 and 0.756 for discovery and optimization, respectively, but prediction MAEs are 0.088 and 0.229. After poor cross-pair experiments, the optimization agent's public prediction rationale assigns low activity to several combinations. Under new conditions it predicts 0.070 [0, 0.240] for one previously tested combination, against a reference of 0.494, while retaining an accurate forecast of 0.570 against 0.572 for another. The records connect experimental evidence, an expressed transfer rule and sealed predictions; they do not show prediction quality changing after each optimization step.

The converse separation also occurs: predictive gains need not accompany better delivery. The 24-batch condition is an independent session with larger experimental and computational allowances (Fig. 4). Electrochemical score MAE improves in 11 of 15 discovery pairs and 12 of 15 optimization pairs. Phase-partitioning organic-fraction MAE improves in 13 of 15 pairs, decreasing from 0.08291 to 0.02379. Crystallization recovery MAE also decreases, from 0.10433 to 0.07937, yet quality-eligible recommendations remain 12 of 15 and mean retested recovery decreases from 0.42054 to 0.39939. Fines-interval coverage rises from 35.6% to 43.9% but remains below nominal 80% in every campaign. More research opportunities can therefore improve selected predictions without improving every delivered outcome.

Relevant scientific information provides another route to improvement. In the reaction parameter study, Aligned information lowers discovery macro MAE from 0.10186 to 0.05701 and optimization error from 0.13893 to 0.08384, outperforming Opaque in four of five worlds under each commission (Supplementary Fig. S2a,b). Across other blocks, no information arm has a common advantage. These gains raise a more specific question: can the value of correct information reverse when prediction moves beyond the conditions that supported it?

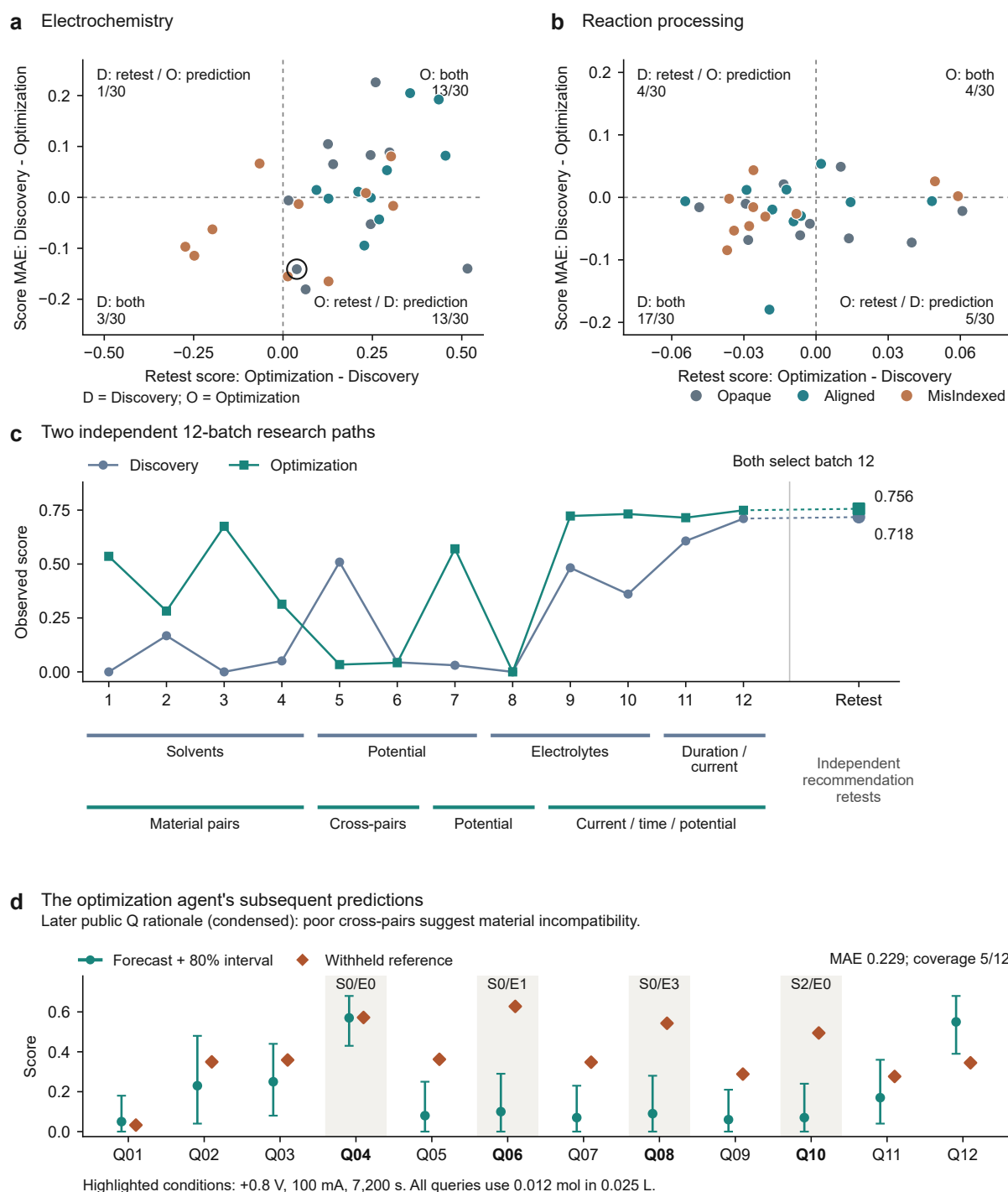


Figure 3: Research objectives, operating outcomes and predictive judgments. a,b, All thirty matched goal pairs in electrochemistry (a) and reaction processing (b), each drawn from five reused worlds. Positive differences favour optimization on the respective endpoint; quadrant labels identify the better goal. Counts use strict signs without a minimum-effect or significance threshold. The ring marks the selected case. c, All batch scores in two independent twelve-batch campaigns, recorded operation changes and independent retests of their sealed recommendations. Lines join successive observations; this is not a twelve-to-twenty-four-batch continuation. Discovery batches 11-12 include segmented electrolysis and intermediate measurement. d, All twelve original optimization forecasts, nominal 80% prediction intervals and withheld reference observations. The rationale condenses a subsequent public Q account, not contemporaneous internal thoughts. Shading identifies four queries at common electrical settings with different material pairs, retaining an accurate forecast alongside underestimates. The case is retrospectively selected. Pairing fixes world, information arm, budget and prior locus. EC references are single seeded observations; RX point errors use five-observation means.

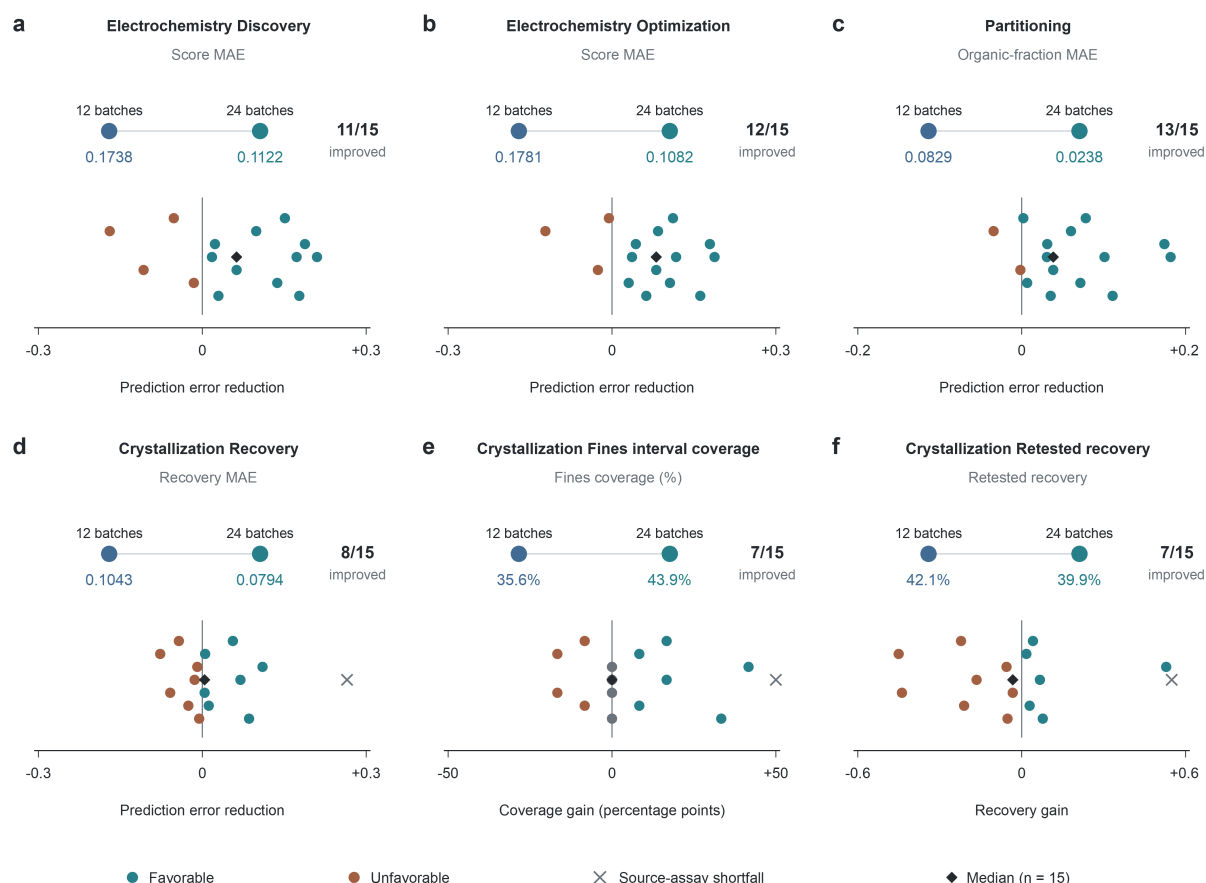


Figure 4: Larger research envelopes have response-specific effects. Panels a-f show electrochemical discovery score MAE, electrochemical optimization score MAE, partitioning organic-fraction MAE, crystallization recovery MAE, crystallization fines-interval coverage and crystallization retested recovery. The upper pair in each panel gives campaign means at 12 and 24 batches and the number of improved world-arm pairs. Each lower plot shows the signed change for all fifteen matched pairs of independent sessions, with vertical offsets used only to separate marks. In a-d, change is MAE at 12 minus MAE at 24; in e and f, change is the 24-batch value minus the 12-batch value. Positive values therefore favour the larger envelope. Black diamonds mark paired medians; grey points at zero mark no change, and crosses retain the C-W02/Opaque pair with a source-assay shortfall. Axes use different scales. The larger batch budget also permits more operational and computational resources. Supplementary Fig. S3 retains the detailed world-arm matrix for all six outcomes.

Correct prior information can reverse its predictive benefit across regimes

The equilibrium study reveals a regime-dependent reversal in the value of correct partial information. Aligned supplies a correct effective-pKa interval and local dilution response, rather than complete model equations (Methods; Supplementary Table H1). Relative to Opaque, it lowers macro MAE from 0.01008 to 0.00465 on nine queries but raises error from 0.01833 to 0.15862 on the three most dilute—approximately 8.7 times Opaque’s error in that group (Fig. 5b; Table 1). Both directions occur in all five worlds. The concentration grouping is post hoc, and the other nine queries include boundary conditions rather than only interpolation.

The acquired evidence does not enter the dilute response regime. All fifteen campaigns sample nominal concentrations above 0.018 M, and dissociation spans 5.5-9.2% across 180 source assays. At the most dilute test, reference means instead span 44.8-71.6% (Fig. 5a). In the executed model, background-ion charge balance produces a local plateau; dilution changes the balance without changing the process laws. One Aligned campaign spans a ninety-fold concentration range and its Opaque counterpart a fourfold range, yet neither enters the dilute regime (Supplementary C). Broad input exploration can therefore leave the transition between response regimes unobserved.

The reversal is response-specific but not confined to dissociation. Aligned’s benefit on the other nine queries comes mainly from precipitation, whereas pH and dissociation have mixed directions; on the dilute three, all three responses worsen in every world (Supplementary Table G1). Its original 80% intervals cover 91.3% of reference observations on the other nine queries but only 2.2% on the dilute three, compared with 90.7% and 93.3% for Opaque. Reported uncertainty therefore does not adequately represent the transfer error.

Table 1. Equilibrium prediction performance across test regimes.

Information arm	Other nine: MAE	Dilute three: MAE	Other nine: coverage	Dilute three: coverage
Opaque	0.01008	0.01833	90.7%	93.3%
Aligned	0.00465	0.15862	91.3%	2.2%
MisIndexed	0.00439	0.14397	89.6%	13.8%

Values average five worlds per arm and three response targets: normalized pH, dissociation fraction and precipitation signal. Coverage concerns the original 80% prediction intervals, with 675 reference-observation judgments per arm for the other nine queries and 225 for the dilute three; these are not independent world replicates. The grouping is exploratory. Figure 5b shows the same macro-error comparison; Fig. 5c isolates one response and query.

At the most dilute condition, Opaque forecasts follow the dissociation increase while Aligned forecasts remain near the observed plateau (Fig. 5c). Public prediction explanations connect this difference to how the evidence is used. Four Opaque accounts invoke piecewise weak-acid extrapolation and the fifth allows departure from the plateau; all five Aligned accounts primarily extend a plateau or weak local trend. MisIndexed retains a counterexample that combines plateau and weak-acid explanations and predicts appreciable dilute dissociation. One Aligned account even acknowledges that the plateau need not extend to zero loading, yet predicts dissociation of 0.0804 [0.053, 0.108] against a reference mean of 0.65463. Expressing an applicability boundary and using it successfully in a sealed prediction are therefore separable achievements. The complementary question is whether forecasts preserve a relationship that remains stable under new conditions.

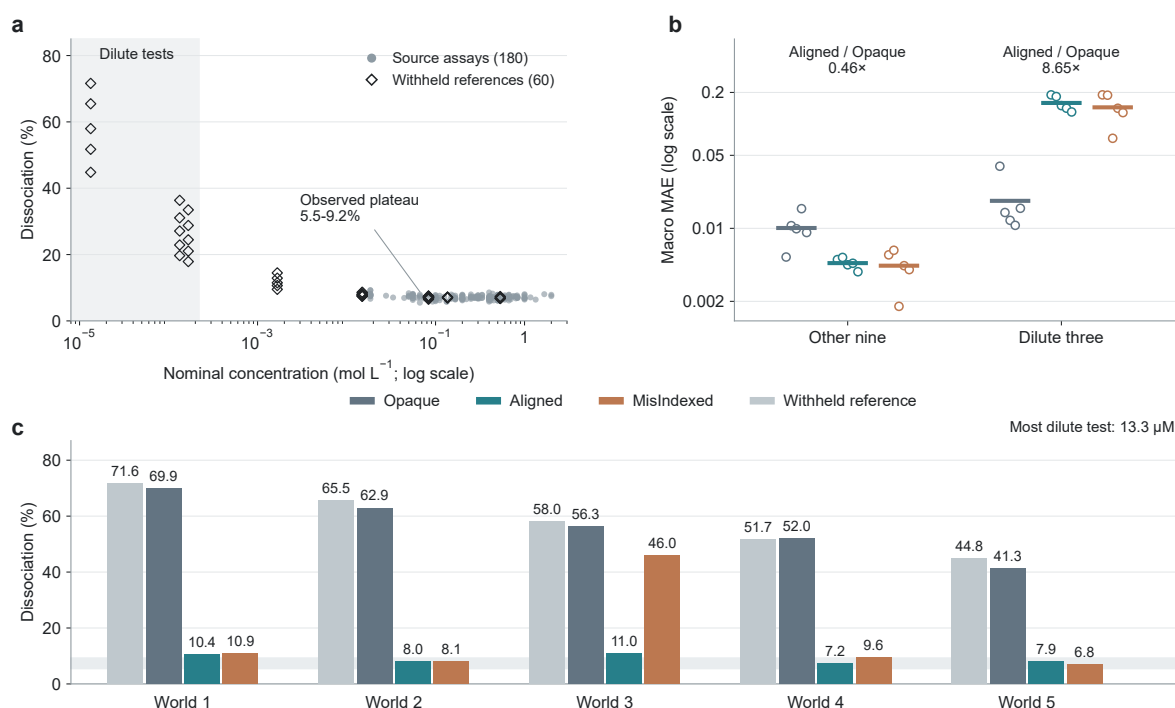


Figure 5: Experimental evidence, regime reversal and original equilibrium forecasts. **a**, All 180 source assays and sixty withheld reference means (five worlds, twelve recipes) plotted against nominal concentration. Diamonds average five evaluator observations; circles are individual source assays. Recipes and histories differ, so the scatter is not a controlled concentration-response curve. Shading identifies the three dilute tests. **b**, Macro MAE for the other nine and three most dilute queries. Open points show all five worlds per arm; short lines show means. Ratios compare Aligned with Opaque within each group. This is the same exploratory grouping and three-response aggregation as Table 1. **c**, Original dissociation forecasts and five-observation reference means for the most dilute recipe, 13.3 micromolar, in all five worlds. The horizontal band spans all source dissociation observations, 5.5-9.2%, and is not an uncertainty interval. World 3 retains the MisIndexed counterexample. All references were withheld from agents; original 80% prediction intervals and campaign-specific source ranges are in Supplementary Table F1.

Useful recovery predictions coexist with systematic purity underestimation

Crystallization exposes the complementary applicability failure: a response remains approximately stable, yet forecasts depart from it. Source and withheld purity average 98.53% and 98.51%, whereas forecasts average 93.54% at 12 batches and 94.89% at 24. Of 360 forecasts, 335 are too low (Fig. 6a), and the mean within-world span of withheld purity is only 2.442 percentage points. The model tracks target and impurity in the solid separately: supersaturation-dependent nucleation and growth alter recovery and particle populations, while modest impurity transfer permits high purity. Its near-constant level is therefore an observed result over the tested conditions, not an imposed constant (Methods; Supplementary H).

The departure is response-specific. A baseline that repeats each campaign's own public final-assay mean uses no additional experiments or withheld outcomes. Agent forecasts outperform it for recovery in 26 of 30 campaigns but for purity in none (Fig. 6b). Mean purity MAE is 4.35 percentage points for the agent versus 0.48 for the baseline. The same researchers can therefore predict changing recovery more effectively than this baseline while failing to preserve near-constant purity. Fines, particle-size and nearest-neighbour comparisons are reported in Supplementary Table E1.

A retained-data decomposition separates a campaign-wide purity offset from error in variation across conditions (Supplementary Table G2). Mean signed bias is -4.29 percentage points, and squared campaign biases account for 87.6% of total squared error. Overall underestimation therefore dominates, but centred RMSE

still exceeds reference standard deviation in 29 of 30 campaigns. Both a systematic level shift and residual errors in variation contribute. This output-level diagnosis does not identify an internal revision process.

Information conditions further constrain the interpretation. At 12 batches, Aligned and MisIndexed improve recovery MAE while worsening purity MAE relative to Opaque in every world; at 24 batches, this joint direction occurs in only two and three of five worlds, respectively (Fig. 6c). Prediction intervals also fail to accommodate the purity errors reliably (Fig. 6d). Aligned and MisIndexed intervals are wider on average than Opaque intervals at both budgets, so poorer coverage cannot be attributed simply to narrower reported uncertainty; conversely, high Opaque coverage alone does not establish calibration or sharpness. The effects depend on both response and research envelope.

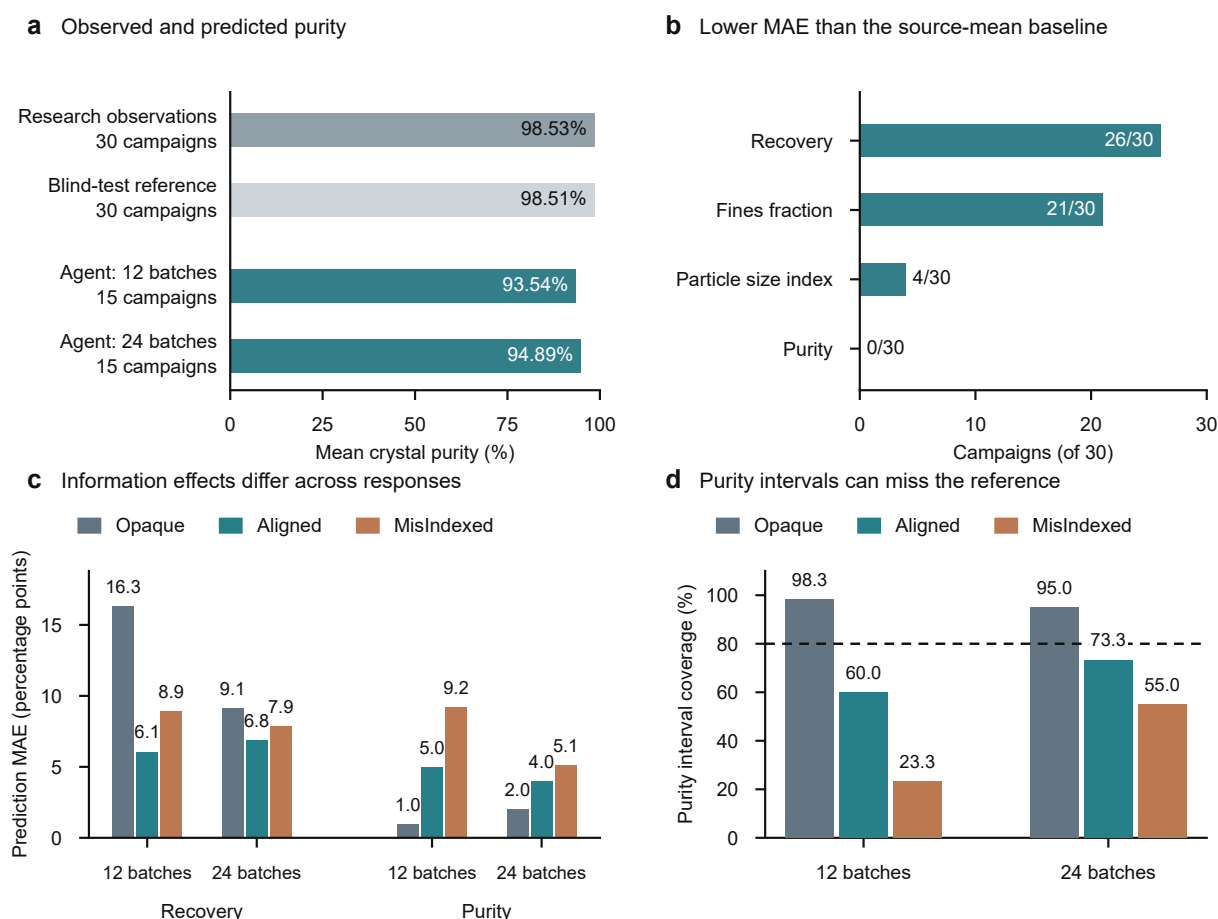


Figure 6: Crystallization forecasts show response-specific departures from experimental evidence. a, Mean purity in source observations and withheld reference outcomes (thirty campaigns each), compared with forecasts at each budget (fifteen campaigns each); bars equally weight campaign means. b, Campaigns with lower agent MAE than a predictor repeating the same campaign’s public final-assay mean for every withheld recipe; there are no ties. Comparisons are within each response, not between response scales. c, Recovery and purity MAE by information arm and budget, with five worlds per bar. d, Coverage of original nominal 80% purity intervals, averaged over five campaigns of twelve queries per arm and budget. The dashed line marks the nominal level. Five worlds recur across arms and independent budget sessions; queries are not independent world replicates. All thirty campaigns, including the source-assay shortfall, remain included. Bars are descriptive summaries, not confidence intervals. The baseline uses no additional experiments or withheld outcomes.

Discussion

Across the tested campaigns, useful procedures and improved predictions did not ensure successful transfer to new conditions. Aligned equilibrium forecasts extended a local dissociation plateau into conditions where the response changed, whereas crystallization forecasts departed from purity that remained approximately stable. Following the original researcher from self-chosen experiments to sealed predictions connects these departures to the evidence acquired during research. Together, the findings motivate applicability judgment—deciding which relationships to preserve and which to revise—as an observable dimension of scientific learning. They do not establish that the two failures share an internal cause.

Evaluation should therefore connect experimental histories, public explanations, predictions and uncertainty. Acknowledging that a relationship has limits does not establish that those limits are reflected in forecasts. Some researchers proposed potentially informative follow-up experiments during retrospective review, after predictions were sealed and without reference feedback, but those experiments remained unexecuted (Supplementary C.4; Supplementary Fig. S4). This suggests a focused next test: whether evidence acquired at a diagnostic boundary improves the original researcher’s subsequent predictions. Such an intervention could test whether acquiring discriminating evidence turns recognition of a boundary into better prediction.

The current evidence concerns one agent configuration, five reused worlds per study and several exploratory comparisons with correlated readouts. It does not isolate the contributions of evidence acquisition and interpretation or establish transfer to physical laboratories. Programmable worlds make these distinctions experimentally accessible while retaining the researcher’s freedom to choose procedures. Further work can test whether the observed patterns recur across agents and whether targeted evidence acquisition changes them. Progress should be assessed through both useful experimental outcomes and predictions that reflect where acquired relationships remain applicable.

Methods

World construction and execution

A world is specified by public experimental contracts and private process parameters. Process components declare state, domains, dependencies and compatible interfaces. Operation and observation models compile these declarations into a laboratory interface. The participant can act through that interface but cannot inspect the simulator, private parameters, evaluator records or repository. Instruments may consume samples and operations consume declared material, time and action allowances. Candidate transitions are validated before commitment; rejection preserves committed physical state while retaining the declared attempt consequences. Replay concerns the environment and recorded operations, not regeneration of language-model responses.

Equilibrium and crystallization process models

The equilibrium parameter study solves monoprotic acid-base speciation with water autoionization and prescribed background-ion inventories, followed by a sequential 1:1 precipitation calculation. Before the ion-amount caps apply, net background charge is 0.07 times the acid amount. Charge balance therefore permits a dissociation plateau near 7% when the hydrogen and hydroxide contributions are small relative to total concentration; their relative contributions change upon dilution. The solver uses current ledger amounts and volume, whereas the reported nominal concentrations describe recipe additions. Supplementary H gives the executed equations and effective-pKa range.

Crystallization couples temperature-dependent solubility to supersaturation-dependent nucleation, growth and impurity transfer. Crystal population and phase inventories persist through cooling, seeding and filtration. Purity is solid target divided by total target-plus-impurity solids. The full-process final-assay recovery removes seed contribution and uses initial reagent charge as denominator. The retained model uses linear supersaturation-dependent impurity incorporation. Near-constant purity is established by these source and withheld outcomes, rather than fixed by the scoring definition. Supplementary H records the constitutive relations and parameters.

Platform qualification

The frozen platform qualification passed 64 task-world units, 1,786 boundary and categorical recipes, 192 invalid-action probes and 52 generated compositions. The generated cases comprise 18 previously absent topologies, eight new task-world identities using a registered topology and 26 additional coverage cases. Component and interface checks passed 32 module probes, seven interface paths and seven invalid compositions. These finite-domain checks establish internal model and execution consistency; they do not establish agreement with arbitrary wet-laboratory chemistry. Exact environment replay reconstructs executed actions and resource consequences, including rejection and rollback.

Qualification distinguishes internal physical and accounting consistency from successful research. It checks executable domains, resources, information separation, valid query references and exact replay. Invalid actions and incompatible compositions have explicit expected rejection or rollback behaviour. Subsequent study adapters have their own checks; the frozen platform qualification is not an automatic certificate for later execution code. The campaign pool combines study blocks collected under their recorded protocols, with differences in reference construction, assessment prompts and recovery handling disclosed below. Post hoc comparisons remain labelled exploratory; this analysis neither changes their execution status nor turns platform checks into validation of scientific understanding.

Study units and objectives

The eight blocks comprise electrochemistry with entity priors (60 campaigns), partitioning with entity priors (30), reaction processing with parameter and structural priors (60), equilibrium with entity, parameter and structural priors (15 each), crystallization with entity priors (30) and purification with entity priors (15). Every block uses five worlds and three information arms. Electrochemistry and reaction processing have discovery and optimization commissions. Electrochemistry, partitioning and crystallization have 12- and 24-batch conditions; other blocks use 12. Parameter-prior P and purification P denote different concepts.

Electrochemistry varies materials, potential, current limits and duration; its optimization objective is a public balanced-efficiency score. Reaction processing varies materials, catalyst loading and thermal histories. Partitioning studies allocation under material, volume and process changes. Equilibrium studies amount, volume, concentration, staged additions, dissociation and precipitation. Crystallization seeks recovery net of seed mass subject to purity at least 0.80, fines fraction at most 0.50 and particles present. Purification seeks original-charge recovery subject to purity at least 0.80. Recovery denominators differ and are not combined.

Agent and information conditions

All source campaigns use GPT-5.6 Sol with medium reasoning effort in independent sessions. Within a block, arms share the physical world, operations, resource contract and withheld questions. The treatment modifies the public dossier according to the block's fixed entity, parameter or structural design. Opaque still exposes the operational interface and general task background. Aligned is a partial nominal description, such as a true effective-pKa interval in the equilibrium parameter study. MisIndexed applies the specified mismatch rather than supplying arbitrary misinformation.

For a concrete equilibrium example, the first world's Aligned input provides an effective-pKa interval of [4.6094, 4.7094] and a local dilution-response claim; MisIndexed provides [5.3294, 5.4294] and the corresponding claim from the fifth world. Both use the same anchor, confidence and qualification as a local relationship; Opaque omits these instance-specific fields. Supplementary Table H1 gives the numerical excerpt reproduced from the frozen input configuration and serializer.

Agents may choose conditions, repetitions, measurement allocations and experiment grouping, and may use the bounded public numerical tool. They are not required to choose from mechanism templates, emit a fixed equation or reflect after each batch. Batches, operations, provider responses and decisions made after feedback are different counts. Each budget permits a matching number of additional measurements and a final assay per completed batch; other resources remain separately bounded. The larger budget includes a larger execution and computation envelope. Retained early electrochemistry and partitioning assessments have an eight-call calculator allowance; later blocks disclose up to 128 attempts. Outputs are requested in English, but retained reaction and equilibrium assessment prompts include Chinese instructions requesting English responses. These differences remain part of the historical protocol.

Sealed assessments and retests

Where required, a completed-batch recommendation is sealed before assessment and its procedure retested independently with a different observation seed in the same world. There are 165 such retests. K1 requests an unrestricted scientific account including relationships, evidence, applicability and unresolved alternatives. Q then requests point predictions and intervals for twelve fixed new-batch recipes without laboratory access or target feedback. K2 follows the sealed predictions with retrospective questions about support, limitations and discriminating experiments. The short version has three questions; reaction and equilibrium use seven themes. The bounded equilibrium parameter block adds fifteen structured supplements.

The original research context is available during every stage. Q can use the source history in addition to K1, and K2 cannot revise Q. Consequently, neither report-only knowledge transfer nor a uniform mechanism-

correctness score is an endpoint. The bounded equilibrium review described below covers all fifteen campaigns in that block; complete mechanism annotation across the 240-campaign programme has not been performed.

Outcome definitions and references

For each response, campaign MAE averages absolute point error across the twelve questions. Interval coverage measures the fraction of specified reference targets contained in the reported intervals; width is their average span. Partitioning uses nominal 90% intervals and the other blocks use 80%. Partitioning and crystallization point references use underlying pre-final-assay response values. Electrochemistry and purification use fixed seeded observations. Reaction and equilibrium point references average five observations; their coverage is evaluated against the individual observations. Macro MAE averages response-specific MAEs equally within the designated block. These targets and scales are not pooled across systems.

The crystallization mean baseline uses only the source campaign's public final assays. Nearest-neighbour prediction additionally uses public recipe features. A parser correction excludes resource-denied attempts from physical recipe features while preserving those attempts in the record. It restores two baseline evaluations; the other twenty-eight remain exactly unchanged. No agent or simulator is rerun for the correction. The equilibrium source-mean diagnostic is a separate post hoc reference. Its failure on dilute queries confirms that copying the plateau is insufficient for those questions; it is not a strong system-identification comparator.

Two additional exploratory diagnostics use only retained predictions and references (Supplementary G). Equilibrium response-specific MAEs retain all 540 point forecasts and reproduce the regime macro means exactly. Crystallization purity error retains all 360 predictions and decomposes each campaign's MSE into squared mean error plus centred MSE. The pooled offset share is the sum of squared campaign biases divided by the sum of campaign MSEs, with equal query counts. Centring uses the withheld references for diagnosis, not to construct a corrected predictor or claim a performance improvement.

Descriptive comparisons and retained failures

Goal comparisons match world, information arm, budget and prior locus. Budget comparisons match world, arm and goal, comparing independent sessions. Arm comparisons match the other factors within each study. Means weight campaigns equally. Five world clusters recur across arms, goals and budgets; questions, response metrics and batches are correlated observations. We report magnitudes, paired directions, all-arm tables and selected sensitivity analyses rather than inferential tests treating queries as independent trials.

The comparison oriented by observed retest performance is a post hoc re-expression of the original goal pairs, not an additional experiment. Within each pair, we identify the campaign with the higher recommendation retest score and compare its score-prediction MAE with that of its counterpart. All pairs and exact effect magnitudes are retained; no minimum-effect threshold is applied. The electrochemical illustration is retrospectively selected and does not estimate the prevalence of its expressed prediction rule. The sensitivity subset excludes any recorded campaign recovery, including assessment-only recovery, rather than source restarts alone.

The equilibrium dilution grouping was selected post hoc using the public concentrations of the three most dilute recipes, approximately 1.33×10^{-4} , 1.33×10^{-5} and 1.67×10^{-4} M. Grouped scores reproduce each original twelve-question macro score when weighted 3:9. The joint recovery-purity crystallization comparison is also exploratory. Neither changes the original query bank or replaces an unfavourable outcome. Retrospectively selected trace examples are distinguished from cohort-level findings.

All 240 campaigns and all 720 scientific-account, prediction and retrospective stages remain included. There are 238 conforming chains and 3,597 of 3,600 planned source assays. One crystallization source has eleven rather than twelve final assays after solvent exhaustion. One purification source consumes twelve starts but has ten final assays and two discarded vessels. Both are marked nonconforming; neither is replaced.

Infrastructure retries retain previous attempts and resource accounting. The supplementary tables report the complete matrix, targets, qualification domains and every response-specific arm mean. Original protocol variation and excluded historical pilots are documented separately from this pool.

Retrospective analysis of the autonomous equilibrium process

We include all fifteen effective parameter-prior equilibrium sources from their retained source and trajectory indexes: five worlds, three arms, 180 batches, 945 operations and 45 canonical assessment stages. Nominal batch concentration is the sum of reagent additions divided by the sum of solvent additions; it is not inferred dissolved concentration. Original question scores and reference observations remain unchanged. No additional model, simulator or judge calls are used.

A single author review assigns descriptive categories to the public Q explanation: piecewise weak-acid extrapolation, qualitative departure from the plateau, plateau or weak-trend extension, and mixed weak-acid/plateau explanation. Exact excerpts and prediction values accompany the classifications; they are not an independent mechanism assessment. K1 evidence is distinguished from hypotheses first explicit in Q. K2 future-experiment choices are classified separately: eleven of fifteen propose trace-load experiments after seeing Q conditions, without receiving reference feedback. These proposals were not executed and do not establish spontaneous discovery or repair during the source campaign. The exported decision-reason fields are unavailable for all 945 operations, so no contemporaneous reasoning sequence is reconstructed from them.

Data and code availability

The source observations, sealed outputs, evaluator-derived summaries and analysis code underlying this draft are retained in the project evidence records. The supplementary material reports aggregate outcomes and protocol details. A public archival package for this integrated study, including the applicable execution version and privacy-reviewed records, has not yet been finalized; the earlier platform release should not be assumed to contain these later studies.

References

- [1] Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624:570–578, 2023. doi: 10.1038/s41586-023-06792-0.
- [2] Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 6:525–535, 2024. doi: 10.1038/s42256-024-00832-8.
- [3] Nathan J. Szymanski et al. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature*, 624:86–91, 2023. doi: 10.1038/s41586-023-06734-w.
- [4] Kobi C. Felton, Jan G. Rittig, and Alexei A. Lapkin. Summit: Benchmarking machine learning methods for reaction optimisation. *Chemistry–Methods*, 1:116–122, 2021. doi: 10.1002/cmtd.202000051.
- [5] Florian Häse, Matteo Aldeghi, Riley J. Hickman, Loïc M. Roch, Melodie Christensen, Elena Liles, Jason E. Hein, and Alán Aspuru-Guzik. Olympus: a benchmarking framework for noisy optimization and experiment planning. *Machine Learning: Science and Technology*, 2:035021, 2021. doi: 10.1088/2632-2153/abedc8.
- [6] Chris Beeler, Sriram Ganapathi Subramanian, Kyle Sprague, et al. ChemGymRL: A customizable interactive framework for reinforcement learning for digital chemistry. *Digital Discovery*, 3:742–758, 2024. doi: 10.1039/D3DD00183K.
- [7] Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Oyvind Tafjord, and Peter Clark. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-0324.
- [8] Kanishk Gandhi, Michael Y. Li, Lyle Goodyear, Agam Bhatia, Louise Li, Aditi Bhaskar, Mohammed Zaman, and Noah D. Goodman. BoxingGym: Benchmarking progress in automated experimental design and model discovery, 2025. Preprint, arXiv:2501.01540.

Supplementary information

A Protocol detail and evidence handling

A.1 Complete study matrix

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
EC / E: electrochemistry	Discovery; score optimization	12, 24	60	1,080 / 1,080
PA / E: phase partitioning	Explain phase allocation	12, 24	30	540 / 540
RX / P and S: reaction and thermal processing	Discovery; safety-constrained optimization	12	60	720 / 720
EQ / E: medium identity	Characterize entity-conditioned responses	12	15	180 / 180
EQ / P: bounded equilibrium	Characterize effective responses	12	15	180 / 180
EQ / S: structural equilibrium	Explain response structure	12	15	180 / 180
C / E: crystallization	Deliver crystals under quality constraints	12, 24	30	539 / 540
P / E: purification	Recover product subject to purity	12	15	178 / 180
Total	Six system families		240	3,597 / 3,600

Table A1. Complete selected study scope. Each campaign has one agent realization. One C campaign exhausts its solvent stock and discards its twelfth vessel, completing eleven final assays. One P campaign uses all twelve vessel starts but discards two vessels, leaving ten final assays. All campaigns have scientific account (K1), sealed prediction (Q), and retrospective review (K2) stages; 238 meet the full source-and-assessment specification.

A.2 Prediction targets and system-specific questions

System	Prediction responses	Reference target	Interval level
EC	Six conversion/efficiency responses, including score	Seeded final observation	80%
PA	Organic and aqueous product fractions	Noiseless pre-sampling fractions	90%
RX	Yield, conversion, selectivity, byproduct signal, risk, score	Five-observation mean; coverage over observations	80%
EQ / E, P and S	Normalized pH, dissociation, precipitation signal	Five-observation mean; coverage over observations	80%
C	Net recovery, purity, size index, fines fraction	Noiseless pre-assay responses	80%
P	Purity, original-charge recovery	Seeded final observation	80%

Table A2. Every campaign predicts twelve conditions. PA fractions are complementary, and its two decision questions reuse the conditions. Reference repeats reduce observation noise but do not add agent sessions.

EC’s twelve conditions balance positive and negative potential while varying duration, materials, and current cap at a stated loading. Its six responses are selective product yield, electrochemical selectivity, Faradaic efficiency, transport efficiency, energy efficiency, and public score. Current is a magnitude cap, not guaranteed delivered current. PA predicts phase fractions before analytical sampling; sampling losses and complementary fractions must not be counted as separate discoveries.

RX’s twelve paths examine time, temperature, catalyst, heating sequence, and quenching. Actual thermal telemetry matters: requested boundary temperature need not be realized reaction temperature. Five reference observations per condition are withheld from agents, with all source assessments sealed before reference-truth release. RX’s 600 reference executions are separate from 720 source batches and 60 retests.

Each EQ block uses twelve questions and five observations per question, giving 300 reference executions. EQ/E crosses three medium identities with three concentrations and a matched-concentration scale control, assessing entity contrast and concentration and scale transfer under a fixed common topology. Bounded EQ/P examines effective aqueous relationships; canonical EQ/S supports response-shape analysis without requiring a named hidden family. An instrument’s equilibrium diagnostic is not the agent’s subjective confidence. The bounded study’s effective-parameter supplement is additional to K1/Q/K2 and does not confer discovery credit for a parameter already supplied as a treatment.

C’s twelve conditions form six pairs: material, seed dose, cooling history, prior thermal history, continued growth, and upstream loading. Its size response is $\min(d_{50}/250 \mu\text{m}, 1)$, a bounded number-weighted index rather than a diameter in micrometres. Fines are particles below 20 micrometres. Quality requires purity and fines criteria jointly, with particles present; recovery of 0.10 is an initial feasibility target, not an optimization ceiling. P’s six paired interventions concern multistage recovery, including wash staging. Its original-charge denominator differs from C’s crystallization-stage recovery.

RX and EQ coverage uses individual reference observations while point targets use means. EC and P use single seeded observations. PA and C prediction coverage concerns noiseless pre-sampling responses; C recommendation retests use observed assays. Apparent retest threshold crossings can therefore reflect noise. Macro errors weight a system’s responses equally; normalized scales do not make responses physically or scientifically interchangeable.

A.3 Language, numerical tools, and stage budgets

Source briefs and requested reports are English. EC/PA and C/P assessment instructions are English; RX and EQ K1/Q/K2 instructions include Chinese text requesting English reports. Within-block arms have the same language contract. Seven-theme K2 reviews cover treatment claims, influential experiments, competing explanations, one proposed discriminating experiment, goal-related tradeoffs, unused evidence and uncertainty, and limitations of the recommendation or scientific account. Three-question versions combine these themes. The reviews occur after source experiments and sealed predictions; their length does not provide additional source experiments.

The earlier EC/PA block retains its saved eight-call diagnostic-calculator hard stop; later blocks disclose 128-attempt assessment allowances. An old EC parameter/structural supplement encountered the smaller limit and remains an execution failure outside the primary E cohort. The repair is not retroactively attributed to older sessions. Public calculation uses agent-accessible information, not hidden targets. Differences in numerical budgets and deadlines are execution contracts, not scientific findings.

A.4 Completion, recovery, and exclusions

The primary denominator is 240 scheduled campaigns, not provider attempts. All 720 K1/Q/K2 stages and 165 planned recommendation retests in EC, RX, C, and P are complete, plus fifteen EQ supplements. Final-assayed source batches total 3,597 of 3,600 planned. C's second-world 12-batch Opaque source has eleven final assays. P's fifth-world Opaque source starts twelve vessels, discards two, and completes ten final assays. Both remain in outcome tables with conformance indicators.

EC/PA has 68 first-attempt and 22 infrastructure-recovered campaigns. Recovery after transport or host failures retains earlier attempts and consumed resources. Missing assessment stages resume from saved context when the source is intact; fresh source replacements are recorded as such. Repeated question delivery and failed-attempt computation are not silently counted as an original clean attempt. EC goal sensitivity is reported with the supplementary goal comparison. First-attempt-only EC-discovery, EC-optimization, and PA budget comparisons improve prediction in four of five, eight of eleven, and eight of ten pairs, respectively; these reduced subsets have uneven coverage.

The current P matrix follows a correction of prepartition inventory handling. Affected qualification and the entire agent block were restarted under the corrected contract; earlier apparently successful cells were not selected into the new matrix. All current source and recommendation trajectories pass inventory, resource, and replay checks. Its two lawful discards are a source-assay shortfall, not an inventory failure. The fixed purity-direction rule treats changes of 0.02 or less as small; a numerical tolerance of 10^{-12} corrects floating-point equality handling. Six direction classifications changed after this correction, with originals preserved. Predictions, point errors, intervals, retests, and denominators were unchanged.

Historical single-world EC parameter/structural supplements, the closed-set EQ/S pilot, early C/P development runs, qualification recipes, references, replays, and interrupted attempts do not enlarge the denominator. Their records remain separate. Frozen platform qualification is also a separate evidence programme. These counts describe execution and analysis scope; they are not 240 independent chemistry replications.

A.5 Baseline correction and trace interpretation

The C empirical references use the same source final observations for all three arms. A recipe parser originally rejected resource-denial transaction records in two sources. The correction preserves those attempts as provenance and excludes them from committed physical recipe features. Recomputing all thirty baselines restores the two missing evaluations and exactly reproduces the twenty-eight previously available results. Original trajectories, truth, model answers, and error records are retained; the correction uses no new simulator or provider calls. Unknown transaction statuses remain errors. This is an evaluator repair, not a change to a completed experiment.

The earlier EC process illustration uses the second-world, 12-batch Opaque discovery campaign; its sparse-positive-coverage counterexamples are first-world, 24-batch Aligned and MisIndexed discovery campaigns. These are distinct from the current article’s Figure 3, which pairs first-world, 12-batch Opaque discovery and optimization campaigns (Supplementary E.6). All are retrospectively selected examples, not a coded prevalence estimate. The additional C case is the fifth-world, 12-batch MisIndexed campaign, chosen as the largest purity MAE in the 12-batch cohort. Its extreme error illustrates a failure; it is not a representative sampling rule.

A systematic analysis should attach each mechanistic claim to its experimental support, contradictory observations, stated domain, and corresponding predictions. It should distinguish an unperformed discriminating experiment, acquired evidence omitted from an account, justified unresolved alternatives, and report-prediction inconsistency. Evaluator knowledge helps identify discriminating interventions; a claim is not wrong merely for using an equivalent representation. Independent review can assist this analysis without replacing measured predictions and retests. No completed cohort-wide mechanism score or causal compression experiment is implied.

A.6 Full prior overview



Figure S1a-d. Full prior overview for electrochemistry across both goals and budgets. Each graphical table retains five worlds and all three information arms; the bottom row is the arm mean. Bar lengths and adjacent values show prediction MAE on the within-panel scale printed above. An x marks a source-assay shortfall. Scales differ between response types. Appendix B retains the full response tables, including the equilibrium entity-prior study.



Figure S1e-h. Full prior overview for partitioning and reaction discovery. Each graphical table retains five worlds and all three information arms; the bottom row is the arm mean. Bar lengths and adjacent values show prediction MAE on the within-panel scale printed above. An x marks a source-assay shortfall. Scales differ between response types. Appendix B retains the full response tables, including the equilibrium entity-prior study.



Figure S1i-l. Full prior overview for reaction optimization and equilibrium. Each graphical table retains five worlds and all three information arms; the bottom row is the arm mean. Bar lengths and adjacent values show prediction MAE on the within-panel scale printed above. An x marks a source-assay shortfall. Scales differ between response types. Appendix B retains the full response tables, including the equilibrium entity-prior study.



Figure S1m-p. Full prior overview for crystallization and purification. Each graphical table retains five worlds and all three information arms; the bottom row is the arm mean. Bar lengths and adjacent values show prediction MAE on the within-panel scale printed above. An x marks a source-assay shortfall. Scales differ between response types. Appendix B retains the full response tables, including the equilibrium entity-prior study.

A.7 Prior effects by system and regime

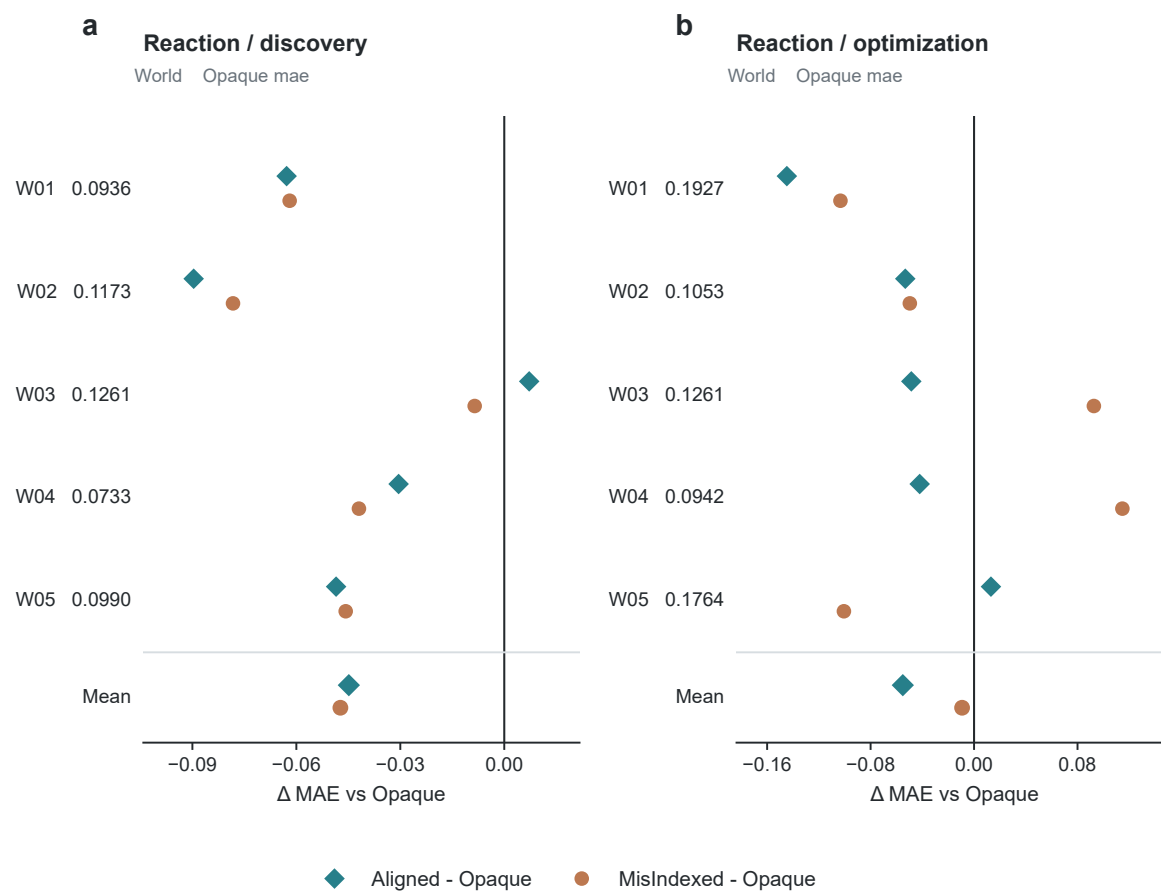


Figure S2a,b. Reaction parameter-prior macro MAE differences from Opaque under discovery and optimization. Rows are matched worlds; the Opaque absolute MAE is printed at left and only paired differences appear on the horizontal axis. Negative means lower error. Bottom marks show the mean paired difference. Aligned improves in four of five worlds under each goal.

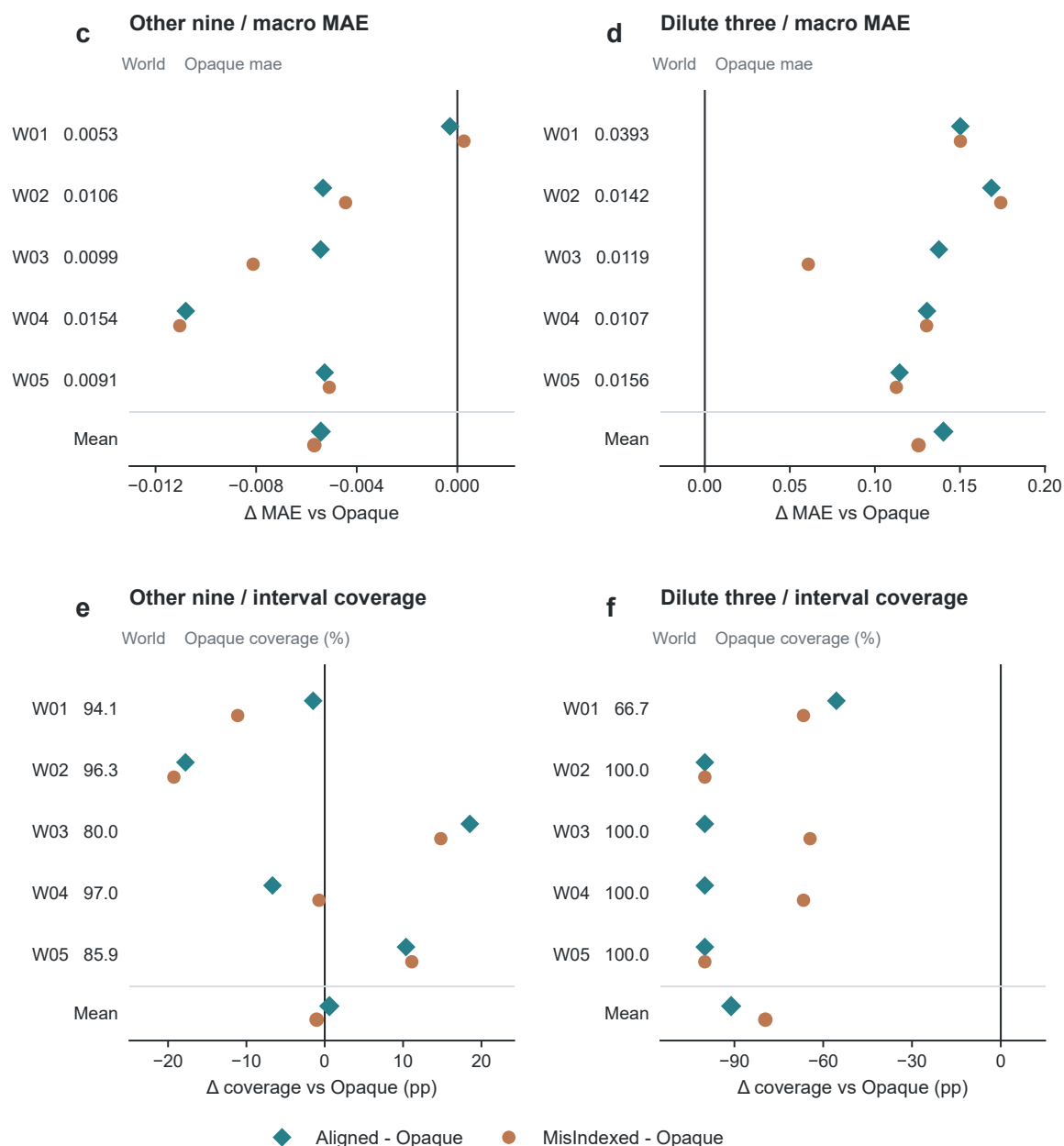


Figure S2c-f. Equilibrium parameter-prior differences from Opaque for the other nine and three most dilute queries. Panels c,d show macro MAE, where negative means lower error. Panels e,f show 80% interval coverage changes in percentage points, where positive means higher coverage. Opaque absolute values are printed at left, apart from the difference axis. Response and regime scales differ. The grouping is post hoc.

A.8 Complete budget comparison

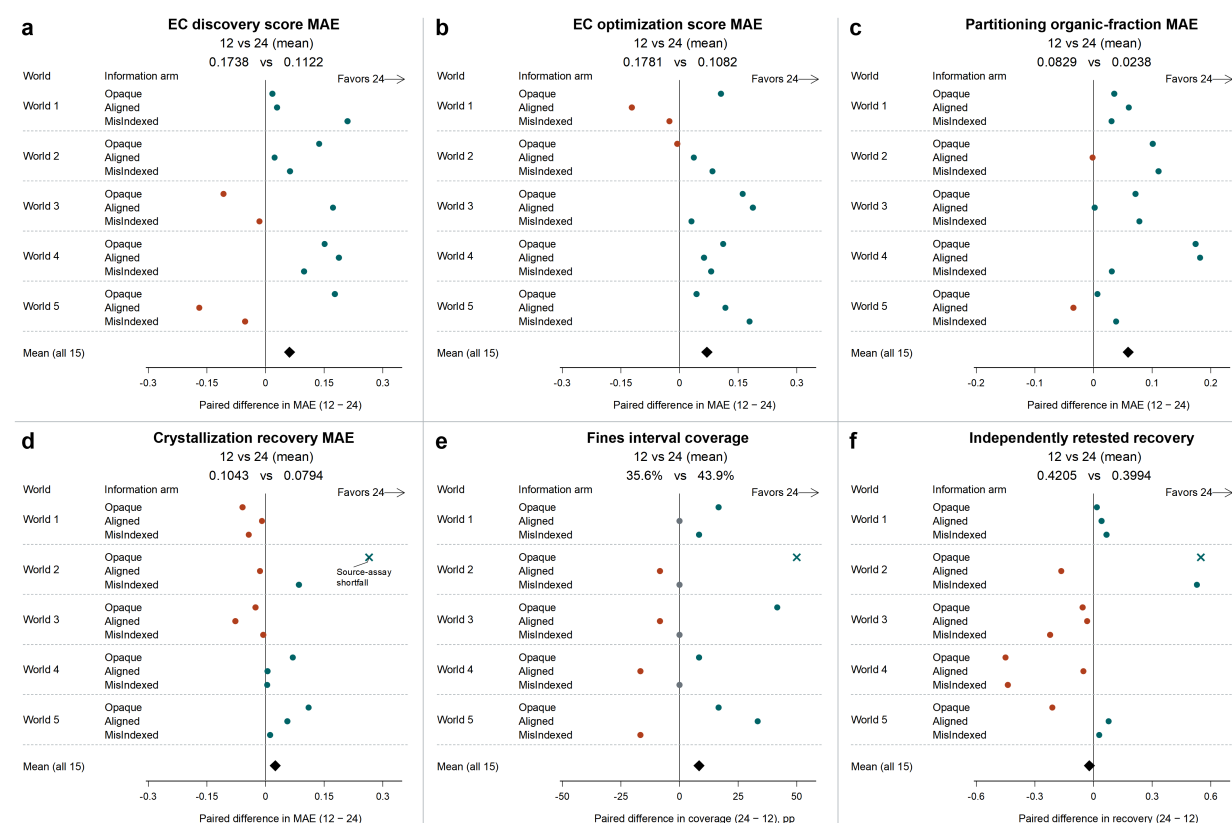


Figure S3: Complete budget effects across six readouts in a two-by-three layout. Each panel retains fifteen pairs of independent sessions. Headers show campaign means; rows show paired changes. Positive changes favour the larger research envelope. Panels a-d show MAE at 12 minus MAE at 24; e shows fines coverage at 24 minus coverage at 12 in percentage points; f shows the corresponding gain in retested recovery. All fines coverage values remain below nominal 80%. Diamonds show mean changes; crosses retain source-assay shortfalls. Marks are point estimates, without uncertainty intervals. World labels are system-specific. Scales differ except in a-b. The crystallization panels reuse the same campaigns; larger budgets also increase computational allowances.

B Complete metric-specific arm means

Each entry is mean absolute error / empirical interval coverage (%), averaged over five world sessions. All scheduled sessions are retained, including the two source-budget shortfalls. PA uses nominal 90% intervals; all other studies use 80%. Metric scales and reference targets differ across systems. The machine-readable table includes each campaign’s interval width and protocol status.

B.1 EC, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2983 / 55.0	0.2740 / 65.0	0.1272 / 88.3
Energy efficiency	0.3098 / 33.3	0.2595 / 51.7	0.1582 / 63.3
Faradaic efficiency	0.2580 / 46.7	0.2259 / 60.0	0.1356 / 73.3
Public score	0.1953 / 51.7	0.1939 / 60.0	0.1321 / 71.7
Selective product yield	0.0731 / 60.0	0.0817 / 38.3	0.0598 / 56.7
Transport efficiency	0.2602 / 45.0	0.2184 / 66.7	0.1328 / 76.7

B.2 EC, E prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2908 / 63.3	0.2378 / 66.7	0.2661 / 73.3
Energy efficiency	0.2794 / 43.3	0.2541 / 31.7	0.2440 / 55.0
Faradaic efficiency	0.2506 / 53.3	0.2532 / 31.7	0.2315 / 56.7
Public score	0.1947 / 55.0	0.1561 / 58.3	0.1835 / 53.3
Selective product yield	0.0758 / 43.3	0.0623 / 53.3	0.0762 / 50.0
Transport efficiency	0.2478 / 58.3	0.2053 / 51.7	0.2217 / 60.0

B.3 PA, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3
Organic fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3

B.4 EC, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1902 / 68.3	0.1268 / 80.0	0.0428 / 93.3
Energy efficiency	0.1581 / 61.7	0.1651 / 68.3	0.0945 / 80.0
Faradaic efficiency	0.1458 / 65.0	0.1505 / 66.7	0.0877 / 78.3
Public score	0.1201 / 63.3	0.1452 / 73.3	0.0713 / 78.3
Selective product yield	0.0726 / 53.3	0.0573 / 70.0	0.0374 / 75.0
Transport efficiency	0.1495 / 65.0	0.1501 / 66.7	0.0846 / 78.3

B.5 EC, E prior: optimization, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1403 / 85.0	0.1252 / 85.0	0.0864 / 90.0
Energy efficiency	0.1977 / 48.3	0.1307 / 60.0	0.1779 / 65.0
Faradaic efficiency	0.1502 / 76.7	0.1098 / 78.3	0.1515 / 71.7
Public score	0.1113 / 75.0	0.0996 / 71.7	0.1136 / 75.0
Selective product yield	0.0515 / 75.0	0.0498 / 56.7	0.0587 / 58.3
Transport efficiency	0.1420 / 80.0	0.1257 / 73.3	0.1570 / 68.3

B.6 PA, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0251 / 100.0	0.0296 / 96.7	0.0207 / 100.0
Organic fraction	0.0211 / 100.0	0.0296 / 96.7	0.0207 / 100.0

B.7 RX, P prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.0800 / 88.7	0.0791 / 73.3	0.0710 / 81.0
Conversion	0.0426 / 72.7	0.0321 / 70.0	0.0306 / 73.3
Safety risk	0.1031 / 58.3	0.0282 / 85.0	0.0370 / 81.7
Public score	0.1195 / 48.7	0.0501 / 83.3	0.0476 / 90.3
Selectivity	0.1198 / 58.0	0.0818 / 69.3	0.0714 / 77.3
Yield	0.1461 / 35.7	0.0708 / 76.7	0.0698 / 78.3

B.8 RX, P prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1509 / 57.7	0.0805 / 73.0	0.1271 / 60.3
Conversion	0.0474 / 85.3	0.0319 / 82.7	0.1960 / 53.0
Safety risk	0.0984 / 56.7	0.1106 / 56.7	0.0811 / 68.3
Public score	0.1341 / 30.0	0.0818 / 73.0	0.0879 / 68.3
Selectivity	0.1900 / 31.0	0.0923 / 69.7	0.1360 / 64.7
Yield	0.2128 / 19.3	0.1060 / 65.7	0.1499 / 68.7

B.9 RX, S prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1219 / 78.7	0.1212 / 75.3	0.1279 / 64.7
Conversion	0.0217 / 97.7	0.0266 / 95.3	0.0200 / 98.0
Safety risk	0.0664 / 80.0	0.0717 / 66.7	0.1276 / 43.3
Public score	0.0504 / 89.7	0.0816 / 61.3	0.0665 / 78.0
Selectivity	0.1280 / 72.7	0.1319 / 70.7	0.1400 / 55.0
Yield	0.1157 / 81.7	0.1196 / 75.7	0.1275 / 61.0

B.10 RX, S prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1376 / 63.7	0.1490 / 67.7	0.1188 / 66.3
Conversion	0.0370 / 88.7	0.0505 / 82.7	0.0425 / 88.3
Safety risk	0.1000 / 45.0	0.1255 / 68.3	0.1133 / 45.0
Public score	0.0932 / 56.0	0.0917 / 58.0	0.0637 / 81.0
Selectivity	0.1574 / 56.7	0.1570 / 60.3	0.1175 / 62.3
Yield	0.1485 / 63.3	0.1422 / 66.3	0.0975 / 73.0

B.11 EQ, P prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0129 / 96.3	0.0759 / 68.7	0.0646 / 74.7
Normalized pH	0.0031 / 94.3	0.0160 / 65.0	0.0150 / 68.7
Precipitation signal	0.0205 / 83.3	0.0376 / 73.3	0.0383 / 68.7

B.12 EQ, S prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0292 / 91.7	0.0235 / 76.3	0.0122 / 87.7
Normalized pH	0.0097 / 87.3	0.0200 / 75.7	0.0203 / 77.7
Precipitation signal	0.0033 / 95.3	0.0042 / 95.7	0.0078 / 87.3

B.13 EQ, E prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0128 / 83.7	0.0107 / 85.0	0.0134 / 79.0
Normalized pH	0.0233 / 80.7	0.0278 / 79.0	0.0202 / 82.0
Precipitation signal	0.0084 / 89.3	0.0116 / 92.0	0.0103 / 88.3

B.14 C, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.3279 / 25.0	0.3096 / 38.3	0.2925 / 43.3
Crystal purity	0.0095 / 98.3	0.0497 / 60.0	0.0917 / 23.3
Particle-size index	0.0777 / 55.0	0.0624 / 58.3	0.0725 / 58.3
Net crystal recovery	0.1632 / 48.3	0.0606 / 83.3	0.0892 / 71.7

B.15 C, E prior: delivery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.2281 / 51.7	0.2774 / 38.3	0.3046 / 41.7
Crystal purity	0.0197 / 95.0	0.0398 / 73.3	0.0508 / 55.0
Particle-size index	0.0566 / 71.7	0.0440 / 70.0	0.1023 / 48.3
Net crystal recovery	0.0911 / 61.7	0.0684 / 80.0	0.0786 / 73.3

B.16 P, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Purity	0.2240 / 51.7	0.1883 / 53.3	0.2508 / 50.0
Recovery	0.0668 / 56.7	0.0687 / 51.7	0.0727 / 53.3

C Exploratory analyses of response regimes

These analyses use the retained source campaigns and their original predictions. They do not define new experimental arms or alter the withheld-condition bank. The concentration grouping, joint crystallization contrast and illustrative traces were selected after the experiments, and are descriptive analyses of reused worlds.

C.1 Equilibrium: a reversal across public concentration regimes

The three most dilute recipes have nominal concentrations approximately 1.33×10^{-4} , 1.33×10^{-5} and 1.67×10^{-4} M. Each campaign has three normalized responses for each recipe. Point errors use five-observation means; interval coverage uses the individual reference observations. Group means average the corresponding responses and queries and then the five world campaigns. Weighting the two groups 3:9 reconstructs every original twelve-question campaign macro MAE. The other nine queries include boundary conditions; this split is not a predeclared interpolation-versus-extrapolation test.

Query group / arm	Macro MAE	Coverage (%)	Interval width	Source-mean MAE
Other nine / Opaque	0.01008	90.7	0.04706	0.00553
Other nine / Aligned	0.00465	91.3	0.03340	0.00518
Other nine / MisIndexed	0.00439	89.6	0.02798	0.00494
Three most dilute / Opaque	0.01833	93.3	0.15676	0.16961
Three most dilute / Aligned	0.15862	2.2	0.08084	0.16833
Three most dilute / MisIndexed	0.14397	13.8	0.09633	0.16836

Table C1. EQ/P subgroup results. Each row averages five worlds; rows reuse those worlds. The source-mean predictor is an additional post hoc diagnostic and obtains no new experiments. Its poor dilute-query performance shows why retaining the plateau is insufficient; its usefulness on the other nine depends on their response distribution.

Aligned improves on the other nine and worsens on the dilute three relative to Opaque in all five worlds. All fifteen source campaigns remain above 0.018 M. In the selected second-world comparison, Aligned spans about ninety-fold source concentration variation and Opaque about fourfold, but both remain in the observed plateau. At the most dilute question, reference dissociation is 0.65463; Opaque predicts 0.629 [0.45, 0.80] and Aligned 0.0804 [0.053, 0.108]. The Opaque prediction response develops an unsaturated weak-acid extrapolation beyond its sealed report. This reasoning must not be credited retrospectively to K1 or to an earlier source-stage hypothesis.

C.2 Crystallization: joint response changes under the same treatment

Budget / arm	Recovery MAE	Purity MAE	Recovery improves and purity worsens vs Opaque
12 / Opaque	0.16322	0.00952	Reference
12 / Aligned	0.06055	0.04969	5 / 5 worlds
12 / MisIndexed	0.08923	0.09166	5 / 5 worlds
24 / Opaque	0.09107	0.01975	Reference
24 / Aligned	0.06839	0.03983	2 / 5 worlds
24 / MisIndexed	0.07863	0.05080	3 / 5 worlds

Table C2. Joint directional comparisons use both response errors from the same world, arm and budget. At twelve batches, each joint direction also holds in all four pairs whose two campaigns conform to the source-assay contract. The direction weakens at 24 batches. A treatment can therefore be helpful on one response and harmful on another without implying a budget-invariant tradeoff.

C.3 Evidence status and interpretation

The same-source C references use only each campaign’s final assays and public recipe features. They exclude hidden truth during fitting but do not reproduce the agent’s complete intermediate measurements, prior or computational policy. Low held-out purity variation makes the mean a particularly strong reference. Neither its success nor the agent’s recovery advantage is a complete mechanism identification result.

Selected EC and C traces support detailed examples rather than population frequencies; the bounded EQ review below covers all fifteen campaigns in that block. The current evidence does not causally identify prior anchoring, report compression, acquisition-versus-inference mediation or a general preference for familiar chemistry. Full-history Q also prevents interpreting K1 as the sole information channel. Experimental selection and later interpretation are both part of the original autonomous process. Their separate causal contributions are not required to describe the observed whole-process contrasts and have not been identified here.

C.4 Original equilibrium campaigns: predictions and research chronology

We reviewed all fifteen retained parameter-prior equilibrium campaigns, comprising 180 batches, 945 operations and 45 K1/Q/K2 stages. Each row below is an original autonomous research session, not an independent reader of its records. The selected question uses 1 micromole in 75 mL; the same question and original reference targets apply to all three arms in a world. The reference column is the five-observation mean.

World / arm	Public Q explanation	Dissociation [80% interval]	Reference
1 / Opaque	Piecewise weak acid	0.6990 [0.250, 0.900]	0.7160
1 / Aligned	Plateau / weak trend	0.1040 [0.044, 0.225]	0.7160
1 / MisIndexed	Plateau / weak trend	0.1092 [0.0772, 0.1412]	0.7160
2 / Opaque	Piecewise weak acid	0.6290 [0.450, 0.800]	0.6546
2 / Aligned	Plateau / weak trend	0.0804 [0.053, 0.108]	0.6546
2 / MisIndexed	Plateau / weak trend	0.0806 [0.0556, 0.1056]	0.6546
3 / Opaque	Piecewise weak acid	0.5630 [0.430, 0.700]	0.5797
3 / Aligned	Plateau / weak trend	0.1100 [0.045, 0.190]	0.5797
3 / MisIndexed	Mixed weak acid / plateau	0.4600 [0.100, 0.750]	0.5797
4 / Opaque	Qualitative departure	0.5200 [0.150, 0.880]	0.5173
4 / Aligned	Plateau / weak trend	0.0724 [0.040, 0.105]	0.5173
4 / MisIndexed	Plateau / weak trend	0.0962 [0.040, 0.450]	0.5173
5 / Opaque	Piecewise weak acid	0.4130 [0.200, 0.700]	0.4480
5 / Aligned	Plateau / weak trend	0.0790 [0.035, 0.130]	0.4480
5 / MisIndexed	Plateau / weak trend	0.0681 [0.040, 0.096]	0.4480

Table C3. Public extrapolation accounts and original same-context predictions. Categories are a single-author retrospective reading of public explanations, not internal reasoning traces or independent causal labels. Four Opaque accounts use piecewise weak-acid reasoning; the fifth permits a qualitative departure. All five Aligned accounts extend a plateau or weak trend. The third-world MisIndexed mixed account is a counterexample to treating all dossier conditions alike.

The first-world Opaque K1 already estimates an effective acid constant near 2.2×10^{-5} and an active-pool cap near 0.004 M, while retaining competing interpretations. Its Q applies that hypothesis at low concentration. In the second world, Opaque K1 primarily supports the observed plateau and limits extrapolation; the unsaturated weak-acid calculation is first explicit in Q. The aligned K1 similarly limits its account to the sampled region, but its Q interval still misses the dilute response by a large margin. These differences separate what was stated before prediction from what became explicit during it.

All fifteen source concentration ranges lie above the three dilute test recipes. The exported per-operation decision-reason fields are unavailable for all 945 operations. Actions and observations can therefore be reconstructed, but a contemporaneous sequence of belief revisions cannot be inferred from these fields. K1, Q and K2 remain distinct public records rather than substitutes for that sequence.

In K2, eleven of fifteen campaigns propose a trace-load experiment: five Opaque, three Aligned and three MisIndexed. The other choices are equal-concentration scale-up, dilution near the supplied relation, comparison of instruments at the same state and repetition of an anomalous batch. K2 follows exposure to Q conditions without feedback on their reference outcomes. None of these proposals was executed; they are retrospective choices, not source-stage plans or demonstrated repairs.

D Selected autonomous research histories

Figure 2 follows one retrospectively selected, twelve-batch MisIndexed crystallization session through a complete series of experiments and its selected operating procedure. Supplementary Fig. S4 presents a separate W05 Aligned 12/24-batch pair and its withheld thermal question. Neither selection estimates a cohort-wide effect; the matched budget comparison is in Fig. 4.

D.1 Complete twelve-batch research path and recommendation

Figure 2 retains all twelve source results in the W04 MisIndexed session. The first four batches have 99.5–100.0% fines and fail the 50% limit. Batch 5 is the first feasible result (40.0% recovery, 30.3% fines) after omitting quench and cooling towards 325 K before seeding. Catalyst, heating, seed and cooling conditions also differ from the early batches; this comparison does not identify a separate quench effect. In the subsequent process family, batches 5–7 end at 275, 265 and 250 K, with recovery 40.0%, 42.8% and 45.0%, respectively. Batch 8 reduces seed from 5 to 1 mg and gives 45.5% recovery and 25.4% fines. Batches 9–10 probe seed omission and longer holding. Batch 11 halves S2 from 0.080 to 0.040 L but yields 44.6% recovery and 72.6% fines, violating quality. Batch 12 repeats the batch-8 recipe and remains feasible (44.1% recovery, 26.9% fines). The exact numerical optimum is less secure than the repeated feasibility of this operating region.

The agent sealed batch 8 as its recommendation. Its source final assay reported 45.5% seed-excluded recovery, 98.2% purity and 25.4% fines; independent retest reported 43.9%, 98.7% and 28.4%, respectively. The recipe charges 0.040 mol reagent, 0.080 L S2 and 0.005 mol C1; heats towards 355 K for 3,600 s at 300 rpm; cools towards 325 K over 7,200 s; adds 1 mg seed; cools towards 250 K over 14,400 s; holds for 14,400 s at 100 rpm; measures particle size, filters, terminates and assays. In this environment, quench stops reaction chemistry while subsequent cooling and crystal growth remain possible. Heating and cooling values are requested targets, not measured temperature trajectories.

The source trajectory records operations and observations but no contemporaneous verbal rationale for each batch. Figure 2c therefore labels decision questions as author reconstructions. K1 was written after all twelve experiments. It interpreted the contrast between quenched and controlled-cooling batches as evidence for nucleation-history dependence while acknowledging that multiple recipe variables changed; the mechanism is plausible but not isolated by this campaign. K2 later identified the confound explicitly. No new experiment was run to distinguish quench from the other changes.

D.2 Sealed predictions and subsequent reflection

Figure S4 follows a separate W05 Aligned pair of independent 12- and 24-batch sessions through sealed prediction and subsequent reflection; the proposed experiment was not executed. In the 24-batch session, the first nineteen batches fail the fines limit, batch 20 first meets it, and batch 23 is selected. The 12-batch session first meets the quality rule at batch 1. These paths and their Q/K2 assessments do not belong to Figure 2's W04 session and are not a matched quench test.

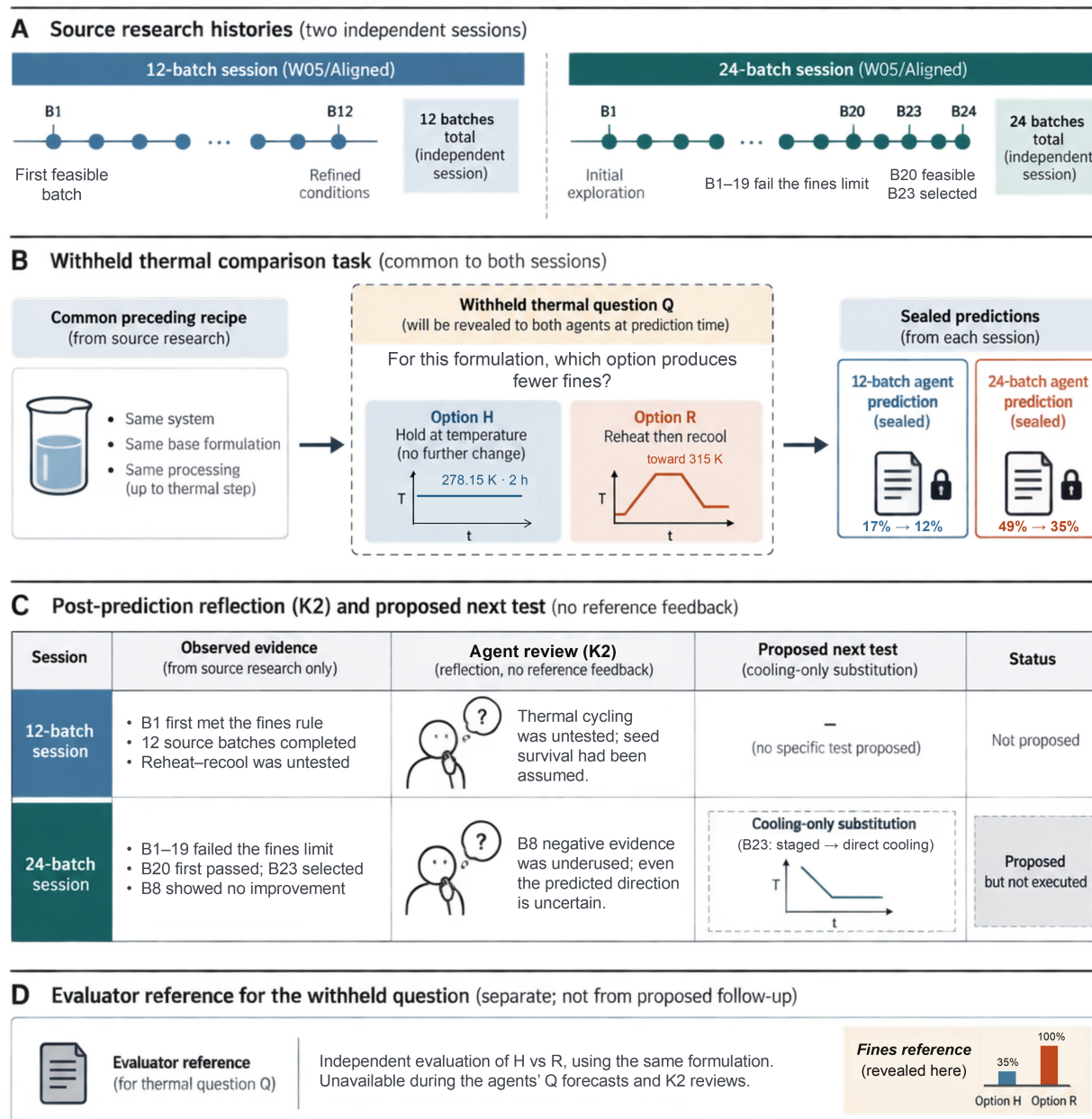


Figure S4: Sealed predictions and subsequent reflection in a W05 crystallization pair separate from Figure 2. A, The independent twelve- and twenty-four-batch source histories: the twelve-batch session first meets the fines rule at batch 1; the twenty-four-batch session first meets it at batch 20 and selects batch 23. B, The same withheld question in both sessions replaces a two-hour hold at 278.15 K with heating towards 315 K for one hour and recooling for one hour after a common preceding recipe. Both sealed Q answers predict fewer fines (17% to 12% at twelve batches; 49% to 35% at twenty-four batches). Thermal profiles show requested targets schematically. C, Condensed public K2 answers follow Q without reference feedback. The twenty-four-batch session proposes changing only the cooling procedure of its selected batch; this experiment was not executed. D, The evaluator's reference is for the thermal intervention in B, not the proposal in C. It was unavailable during Q and K2 and gives the opposite fines direction.

E Supporting outcomes and examples

E.1 Quality constraints and response-specific references

Purification gives a related distinction between recovery and acceptable delivery. Aligned recommendations meet the purity threshold in three of five worlds, compared with none for Opaque or MisIndexed. Yet mean recovered fractions are 0.04455, 0.12373 and 0.32105 for Aligned, Opaque and MisIndexed, respectively. Recovering more material therefore does not imply satisfying the commission. A single retest near the purity threshold is also not a precise estimate of a procedure's reliability. These system-specific outcomes cannot be represented faithfully by a universal yield score (Fig. S5a).

Purification provides another response-specific example: fourteen of fifteen campaigns correctly judge a small concentration effect on purity, but only two correctly judge the small purity effect of splitting a wash. Recognizing approximate invariance is itself a predictive achievement, and depends on the intervention.

In crystallization, agent size forecasts have lower MAE than the source mean in four of thirty campaigns and fines forecasts in twenty-one (Fig. S5b,c). Together with the recovery and purity contrasts in the main text, these results retain the complete response-specific comparison.

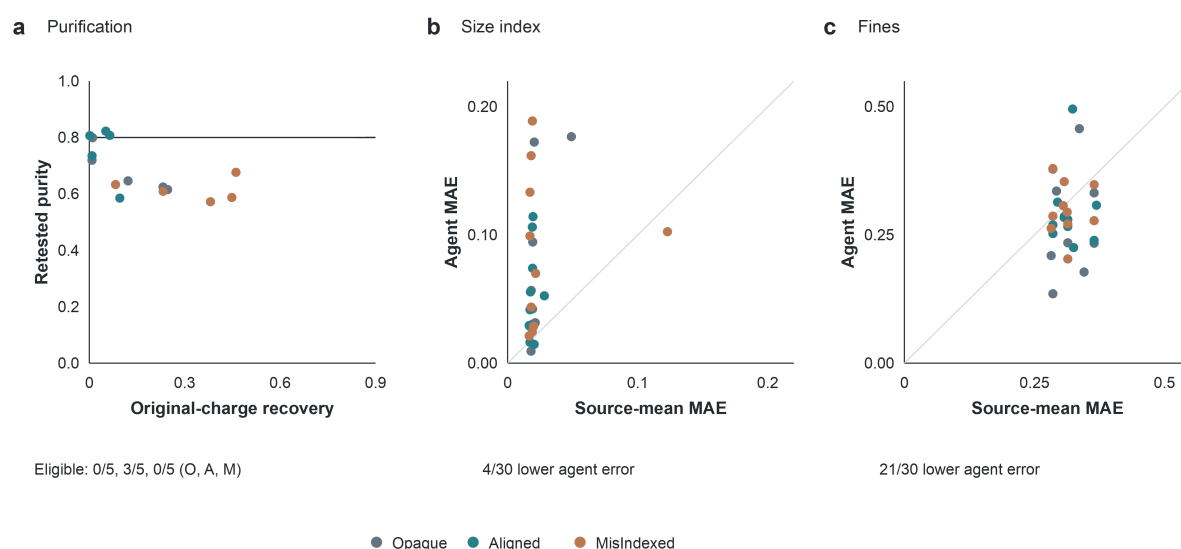


Figure S5: Additional operational and predictive outcomes. a, All fifteen purification recommendations, showing original-charge recovery and independently retested purity. The line marks the purity threshold of 0.80; eligible counts are 0/5, 3/5 and 0/5 for Opaque, Aligned and MisIndexed. b,c, Size and fines MAE for all thirty crystallization campaigns relative to each campaign's public observation mean. Points above the diagonal favour the mean; the campaign with a source-assay shortfall remains included. These panels use different response scales.

E.2 Additional selected examples

In a selected Aligned phase-partitioning campaign pair, the larger budget covers all sixteen material pairings and then allocates eight experiments to process variation; the smaller campaign covers ten pairings. This is a plausible route to improvement, not a cohort-wide causal explanation. Two Aligned world pairs nevertheless worsen.

Uncertainty is part of the same applicability problem. The poor dilute-equilibrium coverage shows that a wrong extrapolation can also be overconfident. In a retrospectively selected MisIndexed crystallization campaign, the scientific account (K1) explicitly acknowledges that only one solvent was tested. Nevertheless, predicted purity averages 0.86833 against a reference mean of 0.98566, and all twelve nominal 80% intervals miss. A verbal statement of uncertainty does not by itself ensure calibrated quantitative prediction.

E.3 Complete crystallization baseline and interval comparisons

Figure 6b reports how often each response is predicted more accurately than the public observation mean. Table E1 supplies the corresponding error magnitudes and the public nearest-neighbour comparison. Each baseline is fitted separately to the same campaign’s public final assays; neither receives withheld outcomes or additional experiments. Nearest-neighbour prediction also uses public recipe features. These are empirical references rather than strong system-identification baselines.

Table E1. Crystallization prediction errors and baseline comparisons.

Response	Agent MAE	Mean MAE	Nearest MAE	Wins: mean	Wins: nearest
Recovery	0.09185	0.17135	0.17308	26/30	26/30
Purity	0.04354	0.00482	0.00905	0/30	4/30
Size index	0.06923	0.02345	0.02708	4/30	7/30
Fines fraction	0.29000	0.31651	0.32426	21/30	20/30

MAE is averaged equally over all thirty campaigns in each response’s original scale; errors are not ranked across responses. Recovery, purity and fines are fractions, while size is the declared size index. Mean and Nearest denote the public observation-mean and nearest-neighbour predictors. Wins count strictly lower agent MAE; neither comparison has ties. The source-assay shortfall remains included, and the five world instances are reused across arms and independent budget sessions.

Table E2 accompanies the purity coverage comparison in Figure 6d. Wider intervals need not cover a biased prediction’s reference, while coverage exceeding the nominal level does not by itself establish calibration or sharpness.

Table E2. Purity interval coverage and width by information arm and budget.

Budget	Information arm	Campaigns	Coverage (%)	Mean width (pp)
12	Opaque	5	98.3	6.81
12	Aligned	5	60.0	12.94
12	MisIndexed	5	23.3	15.53
24	Opaque	5	95.0	8.59
24	Aligned	5	73.3	12.37
24	MisIndexed	5	55.0	12.31

The original intervals have nominal 80% coverage and concern the pre-final-assay response target. Coverage and interval width are averaged over each campaign’s twelve queries and then over its five-world arm-budget group. Each row contains sixty query targets but only five world clusters; queries are not independent world replicates. Width is expressed in percentage points of purity. All original intervals and source failures remain unchanged.

E.4 Complete research-objective comparison

Figure 3 uses 120 campaigns and sixty original goal pairs. Each pair fixes world, information arm, budget and prior locus. EC has five worlds, three arms and two budgets at the entity locus; RX has five worlds, three arms and two loci at twelve batches. The independent recommendation retest and twelve-query score MAE are separate endpoints. EC references are single seeded observations; RX point references are five-observation means. No new campaigns or predictions were produced for this analysis.

Table E3. All joint goal outcomes.

Higher retest	Lower score MAE	EC pairs	RX pairs
Optimization	Optimization	13	4
Optimization	Discovery	13	5
Discovery	Optimization	1	4
Discovery	Discovery	3	17

Both endpoints favour the same campaign in 16/30 EC and 21/30 RX pairs, and opposite campaigns in 14/30 and 9/30. Strict signs define these descriptive counts; near-zero differences are not thereby significant. Each system reuses five worlds. RX score MAE favours optimization in 8/30 pairs; its separate six-response macro MAE does so in 6/30.

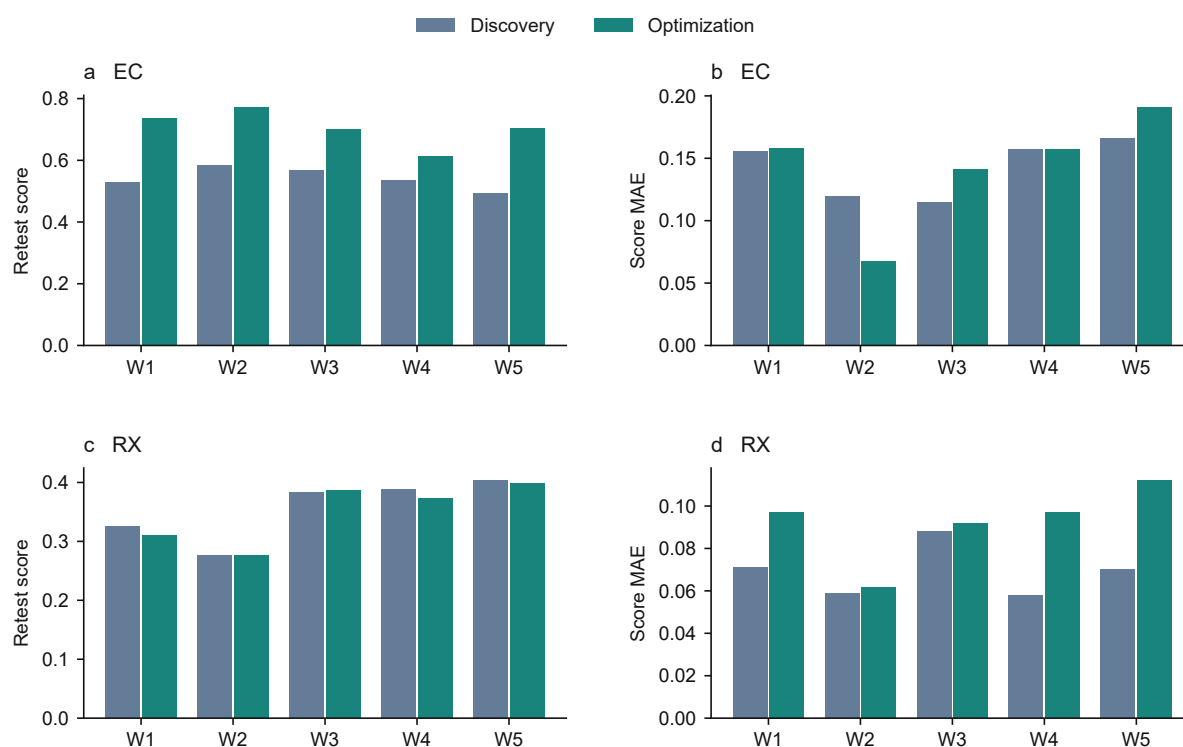


Figure S6: World-level summaries of the same goal comparison. a,b, Electrochemical recommendation retests and score-prediction MAE. c,d, Reaction-processing endpoints. Each bar averages six campaigns per goal within a world. World labels are system-specific; bar means do not replace the individual paired differences in Figure 3. Scales differ across panels.

E.5 Strata and recovery sensitivity

Table E4. Endpoint orderings within design strata.

System	Factor	Level	Pairs	Same	Opposite
EC	World	W01	6	3	3
EC	World	W02	6	4	2
EC	World	W03	6	3	3
EC	World	W04	6	5	1
EC	World	W05	6	1	5
EC	Budget	12	15	7	8
EC	Budget	24	15	9	6
EC	Arm	Aligned	10	6	4
EC	Arm	MisIndexed	10	5	5
EC	Arm	Opaque	10	5	5
EC	Locus	E	30	16	14
RX	World	W01	6	4	2
RX	World	W02	6	4	2
RX	World	W03	6	3	3
RX	World	W04	6	4	2
RX	World	W05	6	6	0
RX	Budget	12	30	21	9
RX	Arm	Aligned	10	6	4
RX	Arm	MisIndexed	10	9	1
RX	Arm	Opaque	10	6	4
RX	Locus	P	15	12	3
RX	Locus	S	15	9	6

Same means that the higher-retest campaign also has lower MAE; opposite means it has higher MAE. These overlapping strata do not add independent tests. Budget rows compare goals at a fixed budget, not a twelve-to-twenty-four-batch contrast. E denotes entity priors, P parameter priors and S structural priors.

The EC subset with neither campaign having a recorded recovery contains seventeen pairs: ten have the same ordering and seven the opposite ordering. Optimization has higher retests in fifteen and lower MAE in eight. This subset excludes assessment-only recovery as well as source recovery; it has uneven coverage and does not replace the full cohort.

E.6 All batches of the selected electrochemical pair

The illustration uses one retrospectively selected twelve-batch Opaque pair in the first electrochemical world. It is not an additional replication or a random sample. Both original agents retain their research context for the later assessments. Every batch uses 0.020 mol in 0.040 L; current denotes the configured cap.

Table E5. Discovery source batches.

Batch	Pair	V	mA	Electrolysis / s	Score
1	S0/E0	-1.5	100	1200	0.000000
2	S1/E0	-1.5	100	1200	0.167530
3	S2/E0	-1.5	100	1200	0.000000
4	S3/E0	-1.5	100	1200	0.050903
5	S0/E0	0.5	100	1200	0.509008
6	S0/E0	1.5	100	1200	0.044293
7	S0/E0	2.5	100	1200	0.030644
8	S0/E1	0.5	100	1200	0.000000
9	S0/E2	0.5	100	1200	0.482517
10	S0/E3	0.5	100	1200	0.360679
11	S0/E2	0.5	100	1200+13200	0.606942
12	S0/E2	0.5	500	1200+13200	0.710658

Table E6. Optimization source batches.

Batch	Pair	V	mA	Electrolysis / s	Score
1	S0/E0	1	100	3600	0.535704
2	S1/E1	-1.5	300	7200	0.282124
3	S2/E2	1.5	300	7200	0.674932
4	S3/E3	1.5	300	7200	0.313717
5	S2/E0	1.5	300	7200	0.033808
6	S0/E2	1.5	300	7200	0.042443
7	S2/E2	1	300	7200	0.570485
8	S2/E2	2	300	7200	0.000000
9	S2/E2	1.5	500	7200	0.722737
10	S2/E2	1.5	500	10000	0.731956
11	S2/E2	1.4	500	9000	0.714786
12	S2/E2	1.5	400	12000	0.749194

Both campaigns select batch 12. Independent retests are 0.717587 for discovery and 0.756122 for optimization; twelve-query score MAEs are 0.088394 and 0.229397.

E.7 Operation sequence and subsequent public accounts

All batches add reagent, add solvent, set potential/electrolyte/current, electrolyze, measure UV-visible response, terminate and obtain a final assay, in that order. Discovery batches 11 and 12 insert a second 13,200-s electrolysis after the first 1,200-s electrolysis and its UV-visible measurement. Their 14,400-s totals therefore do not describe one uninterrupted operation. The records contain 86 discovery and 84 optimization operations. The phase labels in Figure 3 summarize recorded choices, not contemporaneous internal thoughts.

The following original public rationales accompany the sealed Q forecasts, after research and K1. They were issued without reference feedback. They are public accounts, not validated mechanisms or evidence that the acquired observations were sufficient.

Discovery Q rationale (verbatim).

Predictions use the campaign's empirical potential-window and charge-delivery trends at nearly the same concentration: +0.8 V is treated as closer to the productive +0.5 V regime than -0.8 V, duration raises conversion and yield, and 500 mA raises yield while penalizing faradaic, transport, and energy efficiencies. Electrolyte and solvent adjustments follow the matched short-run screens. Intervals are deliberately broad because both test potentials are unmeasured, long-time data exist only at +0.5 V/E2, and low-yield assay noise strongly affects short runs and the score's yield gate.

Optimization Q rationale (verbatim).

Predictions use the observed S0/E0 batch as the main local anchor, scale product formation with electrolysis duration and configured current, and impose the campaign's strong positive-potential window. The S0/E1, S0/E3, and S2/E0 combinations are assigned low activity because cross-pair controls showed severe nonadditive solvent-electrolyte incompatibility. Higher current raises expected yield but lowers transport and charge efficiency. Intervals include final-assay noise plus substantial extrapolation uncertainty, especially for negative polarity and untested material pairs.

E.8 Complete optimization forecasts

All twelve forecasts use 0.012 mol in 0.025 L. Predictions and nominal 80% intervals were sealed before reference feedback. References below are the original seeded observations. The four bold query identifiers share +0.8 V, 100 mA and 7,200 s, while changing the material pair.

Table E7. Query settings, sealed forecasts and reference scores.

Q	Pair	V	mA	s	Forecast	80% interval	Reference
01	S0/E0	-0.8	100	600	0.050	0.000-0.180	0.032659
02	S0/E0	0.8	100	600	0.230	0.040-0.480	0.349491
03	S0/E0	-0.8	100	7200	0.250	0.080-0.440	0.358985
04	S0/E0	0.8	100	7200	0.570	0.430-0.680	0.572012
05	S0/E1	-0.8	100	7200	0.080	0.000-0.250	0.362359
06	S0/E1	0.8	100	7200	0.100	0.000-0.290	0.627599
07	S0/E3	-0.8	100	7200	0.070	0.000-0.230	0.347853
08	S0/E3	0.8	100	7200	0.090	0.000-0.280	0.542955
09	S2/E0	-0.8	100	7200	0.060	0.000-0.210	0.288226
10	S2/E0	0.8	100	7200	0.070	0.000-0.240	0.494376
11	S0/E0	-0.8	500	7200	0.170	0.040-0.360	0.276451
12	S0/E0	0.8	500	7200	0.550	0.390-0.680	0.344881

Score MAE is 0.229397; interval coverage is 5/12. The accurate Q04 prediction is retained alongside the large underestimates at Q06, Q08 and Q10. Source-to-query changes involve amount, volume and electrical settings, so these contrasts do not isolate one changed variable. Twelve query outcomes do not establish population calibration. No intermediate sealed prediction checkpoints were collected, so final prediction errors cannot be plotted as a within-session learning curve.

F Complete most-dilute equilibrium predictions

F.1 Original intervals and campaign-specific observations

The main-text equilibrium comparison shows point predictions for the most dilute recipe. Table F1 retains all fifteen original 80% prediction intervals and each campaign’s source-response range. These are prediction intervals issued by the agent, not confidence intervals on an estimated mean. Each source range covers twelve final assays; the reference is the mean of the original five evaluator observations and was unavailable during prediction or retrospective reflection.

Table F1. Dissociation predictions at the most dilute recipe, with all values in percent.

World / arm	Source range	Point prediction	80% interval	Reference mean
1 / Opaque	6.18-8.34	69.90	[25.00, 90.00]	71.60
1 / Aligned	6.20-9.23	10.40	[4.40, 22.50]	71.60
1 / MisIndexed	6.11-9.25	10.92	[7.72, 14.12]	71.60
2 / Opaque	6.17-8.57	62.90	[45.00, 80.00]	65.46
2 / Aligned	6.12-8.06	8.04	[5.30, 10.80]	65.46
2 / MisIndexed	6.23-7.98	8.06	[5.56, 10.56]	65.46
3 / Opaque	5.94-8.22	56.30	[43.00, 70.00]	57.97
3 / Aligned	6.34-8.02	11.00	[4.50, 19.00]	57.97
3 / MisIndexed	6.16-7.94	46.00	[10.00, 75.00]	57.97
4 / Opaque	6.19-7.91	52.00	[15.00, 88.00]	51.73
4 / Aligned	5.82-8.14	7.24	[4.00, 10.50]	51.73
4 / MisIndexed	5.81-8.22	9.62	[4.00, 45.00]	51.73
5 / Opaque	5.49-7.48	41.30	[20.00, 70.00]	44.80
5 / Aligned	6.44-7.99	7.90	[3.50, 13.00]	44.80
5 / MisIndexed	6.58-7.79	6.81	[4.00, 9.60]	44.80

The recipe contains 1 micromole in 75 millilitres (nominal concentration 13.3 micromolar). The pooled research-observation band in the main figure is the minimum and maximum across all 180 assays; the ranges above are specific to each campaign. World 3 MisIndexed retains its mixed-explanation departure from the observed plateau. Public explanation categories and K1/Q chronology remain in section C.4.

G Response contributions and purity-error diagnosis

G.1 Equilibrium reversal by response

This exploratory decomposition retains all fifteen parameter-prior campaigns and all 540 response predictions. It uses the existing three-lowest-concentration grouping. MAE first averages queries within a campaign and response, then the five worlds. Averaging the three response MAEs exactly recovers Table 1. The same worlds recur across arms; queries and responses are not independent replicates.

Table G1. Original prediction MAE by response and concentration group.

Response	Group	Opaque	Aligned	MisIndexed	Lower / higher
pH / 14	Other nine	0.00261	0.00269	0.00254	2 / 3 of 5
pH / 14	Dilute three	0.00448	0.05576	0.05230	0 / 5 of 5
Dissociation fraction	Other nine	0.00839	0.00709	0.00660	3 / 2 of 5
Dissociation fraction	Dilute three	0.02640	0.28229	0.23861	0 / 5 of 5
Precipitation signal	Other nine	0.01923	0.00417	0.00403	5 / 0 of 5
Precipitation signal	Dilute three	0.02411	0.13783	0.14099	0 / 5 of 5

The final column counts worlds with lower/higher Aligned MAE than Opaque. Signed response differences divided by three sum to the macro-error difference. They quantify the contribution on the declared normalized scales; they do not establish a shared failure mechanism or physical comparability of targets.

G.2 Separating purity offset from variation error

For each of thirty campaigns, let p_i and y_i be its original prediction and retained noiseless reference for the twelve queries. Define $b = \overline{p} - \overline{y}$ and $c_i = (p_i - \overline{p}) - (y_i - \overline{y})$. Then

$$\text{MSE} = b^2 + \frac{1}{12} \sum_{i=1}^{12} c_i^2.$$

The first term measures the common offset; the second measures mismatched variation across conditions. Centred RMSE is the square root of the second term. Reference and prediction standard deviations use denominator twelve. All calculations retain the source-assay shortfall. Demeaning uses withheld references only for diagnosis; it is not an available predictor or a corrected performance result. MAE has no analogous additive decomposition.

Table G2. Purity offset and variation diagnostics.

Batches	Arm	N	Bias	cRMSE	Ref. SD	Pred. SD	Offset share
Both	All	30	-4.289	1.889	0.598	1.968	87.6%
12	Opaque	5	-0.806	0.970	0.598	1.031	45.3%
12	Aligned	5	-4.920	1.837	0.598	1.791	89.8%
12	MisIndexed	5	-9.166	2.843	0.598	2.977	91.3%
24	Opaque	5	-1.816	1.682	0.598	1.649	57.4%
24	Aligned	5	-3.948	1.797	0.598	1.983	85.6%
24	MisIndexed	5	-5.080	2.206	0.598	2.376	85.3%

N is the campaign count; cRMSE denotes centred RMSE. Bias, cRMSE and both standard deviations are campaign means in percentage points. Offset share is the ratio of summed squared campaign biases to summed campaign MSEs, with equal query counts; it is not the mean campaign fraction. The median campaign fraction is 77.6%. Mean bias is negative in 30 of 30 campaigns. Centred RMSE exceeds reference SD in 29 of 30. These comparisons describe forecast outputs and do not identify an internal revision process. No new agent or simulator calls were made.

H Executed process models and information conditions

H.1 Equilibrium response model

The parameter-prior study uses a bounded monoprotic acid-base model followed by a precipitation calculation. These are the process relations used for the retained campaigns and references. They are distinct from explanations proposed by the agents. In particular, an agent’s proposed active-pool cap is not the implemented source of the dissociation plateau.

Let n be the current ledger amount of the initial reactant, V the current sample volume and $C = n/V$. The background-ion amounts are $n_+ = \min(0.020, 0.15n)$ and $n_- = \min(0.020, 0.08n)$, in moles. With $H = [\text{H}^+]$, $K_a = 10^{-\text{p}K_a}$ and activity-ratio factor g , the solver obtains

$$[\text{A}^-] = \frac{CK_a}{K_a + gH}, \quad [\text{OH}^-] = \frac{K_w(T)}{H},$$

subject to charge balance,

$$H + \frac{n_+}{V} - \frac{K_w(T)}{H} - [\text{A}^-] - \frac{n_-}{V} = 0.$$

The reported responses are $\text{pH}/14 = -\log_{10}(H)/14$ and dissociation fraction $\alpha = [\text{A}^-]/C$, clipped to the public unit interval. In the uncapped background-ion range, charge balance implies $\alpha = 0.07 + H/C - K_w/(HC)$. At sufficiently high concentration the last two terms are small, giving a local plateau near 7%; dilution makes their contribution appreciable. The plateau and departure therefore arise under the same fixed equations. The five effective $\text{p}K_a$ values span 4.6594 to 5.3794 in increments of 0.18; the default activity-ratio factor is one.

Precipitation is calculated sequentially from effective ion inventories $a = n_+ + 0.10n_{\text{A}^-}$ and $b = n_- + 0.08n_{\text{HA}}$. When supersaturated, a 1:1 removal x , bounded by the limiting inventory, satisfies $(a - x)(b - x)/V^2 = K_{sp}$, with default $K_{sp} = 1.8 \times 10^{-10}$. The public signal is x/n , clipped to $[0, 1]$. This hook does not feed precipitation back into the acid-base solution. Thus the three public responses are related but should not be interpreted as three independent measurements of one failure mechanism.

The concentration in Figure 5 is a recipe descriptor: total nominal reagent additions divided by total solvent additions. The solver instead uses the ledger state at measurement, after any sample consumption and other operations. The scatterplot combines distinct recipes and histories; it is not a controlled one-dimensional titration curve. The three most dilute test recipes have nominal concentrations of approximately 1.33×10^{-4} , 1.33×10^{-5} and 1.67×10^{-4} M. No source assay in these fifteen campaigns reaches those concentrations.

H.2 A concrete three-arm information example

Table H1 reproduces the numerical content of the first equilibrium world’s public prior from the frozen configuration and input serializer. It is a formatted input excerpt, not a reconstruction of agent dialogue. All three arms retain the same public operations, general task background, resource contract and physical world. Opaque receives no instance-specific initial world model. The other two arms receive the same schema and wording; MisIndexed substitutes the configured fifth world’s claim.

Table H1. Public prior content in one matched physical world.

Supplied field	Opaque	Aligned	MisIndexed
Effective pKa interval (80%)	Absent	[4.6094, 4.7094]	[5.3294, 5.4294]
Dilution change in pH/14	Absent	+0.003494987	+0.000875889
Dilution change in dissociation	Absent	+0.008187416	+0.001884676
Nominal confidence	Absent	0.8	0.8

The common dilution anchor is 0.001 mol acid at 298.15 K, with volume increasing from 0.018 to 0.054 L. Both supplied records identify the source as an “independent bounded archival fit” and qualify the claim as a “local effective relationship; not a universal aqueous-chemistry law”. The pKa intervals above are rounded for display; the input contains their full precision. Aligned therefore supplies correct partial local information, not the evaluator’s complete process equations, query answers or an instruction to enter the most dilute test region. The experiment contrasts the resulting complete research processes, including possible changes to evidence acquisition and interpretation.

H.3 Crystallization response model

The retained crystallization block composes upstream reaction, thermal operations, seeding, crystal nucleation and growth, filtration and measurement. It uses the latent-material process family and its linear supersaturation-dependent impurity transfer. Solid product, solid impurity and the particle population persist across operations; dissolved and solid inventories are accounted for separately.

The temperature-dependent solubility follows

$$c^*(T) = c_{\text{ref}}^* \exp \left[-\frac{\Delta H}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right],$$

with $T_{\text{ref}} = 298.15$ K, $\Delta H = 20,000$ J mol⁻¹ and a declared temperature domain of 250-430 K. World and solvent properties scale the reference solubility and kinetics. For relative supersaturation $s = \max(C/c^* - 1, 0)$, primary nucleation scales as $B = k_b s^2$ and linear growth as $G = k_g s$. Before world and solvent multipliers, the coefficients are 2×10^7 L⁻¹ s⁻¹ for nucleation and 2×10^{-8} m s⁻¹ for growth. Nucleation and growth consume available dissolved target and update the population distribution; they are not independent score adjustments.

Each crystallized target increment transfers impurity according to

$$\Delta n_{I,s} = \min(n_{I,l}, \Delta n_{P,s} k_{\text{occ}}(1 + 0.5s)),$$

where $k_{\text{occ}} = 0.02$ times the solvent's occlusion multiplier. The four nominal solvent multipliers are 1.20, 0.80, 0.65 and 1.05; world-fixed material residuals further modulate these properties. Filtration retains 96% of target solids and 92% of impurity solids. The final-assay purity is the solid-target mole fraction $n_{P,s}/(n_{P,s} + n_{I,s})$. Seed contributes to target purity but is subtracted in the full-process recovery numerator; final-assay recovery is normalized by the initial reagent charge.

These relations allow thermal and seeding histories to alter the amount and size distribution of crystals while impurity incorporation remains modest. They explain how recovery and particle responses can vary without requiring a comparable purity change. The approximately 98.5% purity level is an empirical property of the retained source and query conditions, not a hard-coded value or a universal invariant of the model. The present analysis does not isolate individual process contributions to that stability. Appendix G separately tests whether the forecasts preserve the level and variation of the retained purity references.

H.4 Scope of this reconstruction

The equations and input excerpt were checked against the execution-version source and retained configuration bindings. Later equilibrium adapters and alternative crystallization impurity laws are not used to explain these observations. This documentation and the error decompositions use existing source records, predictions and references only; they add no agent or simulator calls. Internal source identities are retained with the analysis records for the archival package.